

Xsuite Imperfections Model with Corrections and Beam-Beam Effects

228th FCC-ee Accelerator Design Meeting – 20th August 2026

Lara Sophie Sievert & Tirsi Prebibaj

lara.sievert23@imperial.ac.uk

Supervisors: T. Pieloni, M. Seidel

Many thanks to:

K. André, S. Jagabathuni, W. Herr, G. Katsanevakis, P. Raimondi, K. Skoufaris, L. van Riesen-Haupt, Y. Wu

Introduction / Motivation

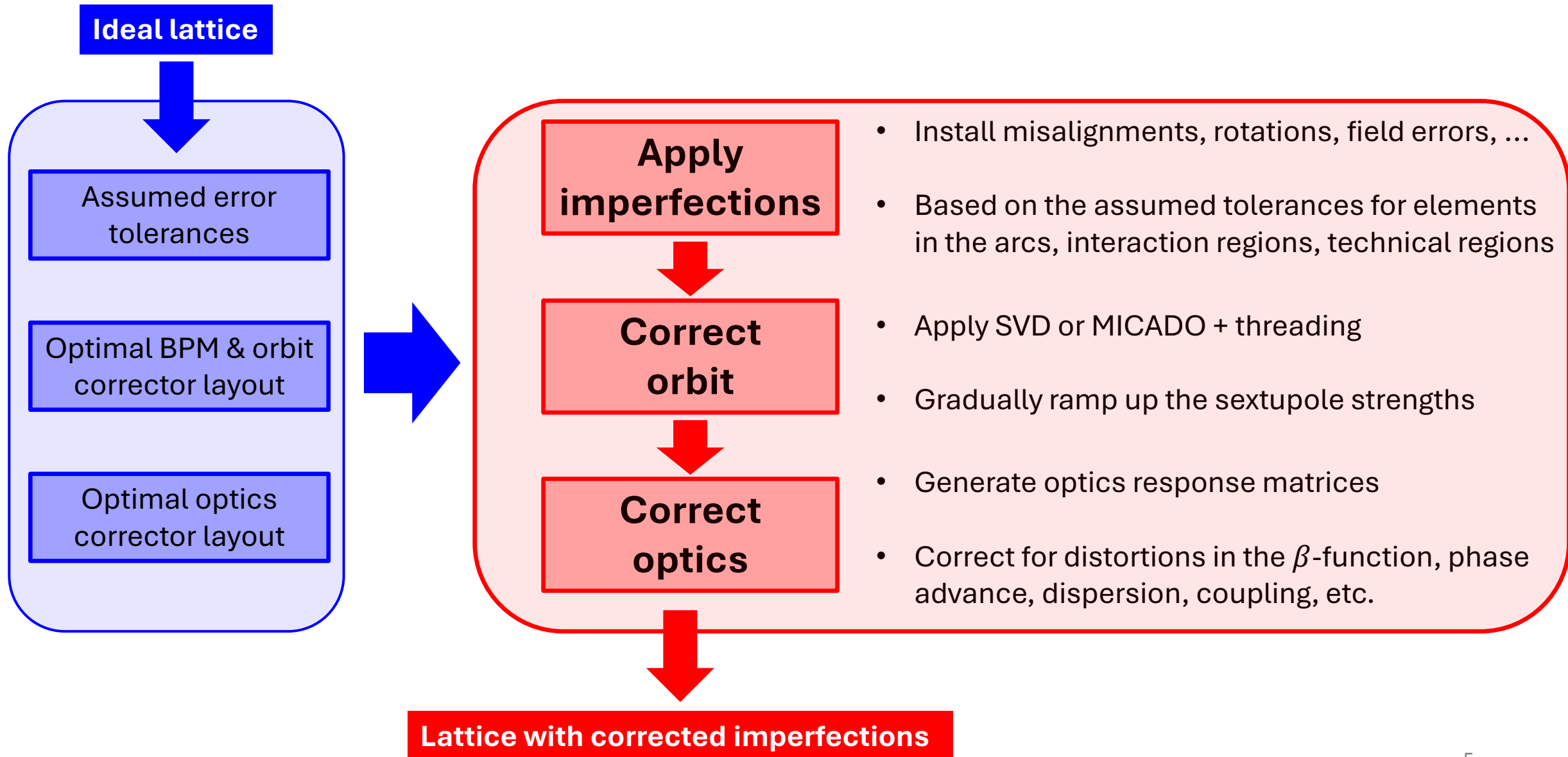
- Master's Thesis for Imperial College London during exchange year at EPFL, in LPAP team since February 2026
 - “**Imperfections Model of the FCC-ee with Corrections and Beam-Beam Effects**” (July 2026)
- Goal: study **interplay between imperfections and beam-beam** in FCC-ee
 - This required imperfections + corrections + beam-beam in one single framework → Xsuite (LCC at Z)
- **Starting point:** imperfections & orbit correction by Y. Wu in MAD-X, reimplemented & extended in Xsuite
 - Several models being developed at the time for GHC (Satya et al.) and LCC (Georgios et al.), and in pyAT (Kevin et al.)
 - Convergence towards a common model now being discussed
- Tried to create a simple, transparent, and generic model, extendable to other FCC-ee energies and to other machines (as will be shown)
 - FCC-ee (LCC at Z) acted as the workhorse for our beam-beam + polarisation studies, as discussed in this presentation

Outline

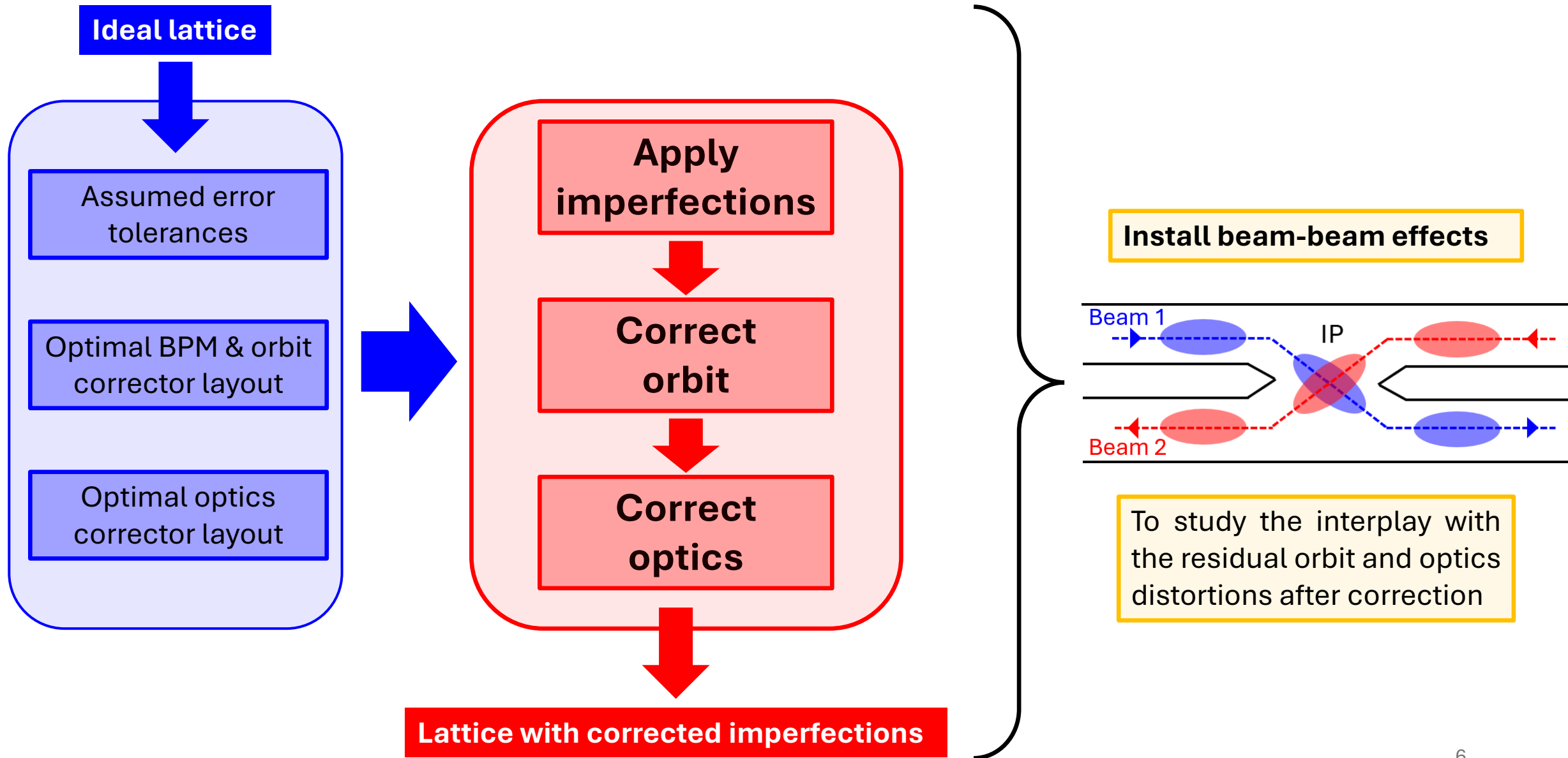
- I. Model Implementation**
- II. Corrected FCC-ee Lattices**
- III. Beam-Beam and Other Studies**

I. Model Implementation

1. How to turn the ideal lattice into a tuned lattice



1. How to turn the ideal lattice into a tuned lattice

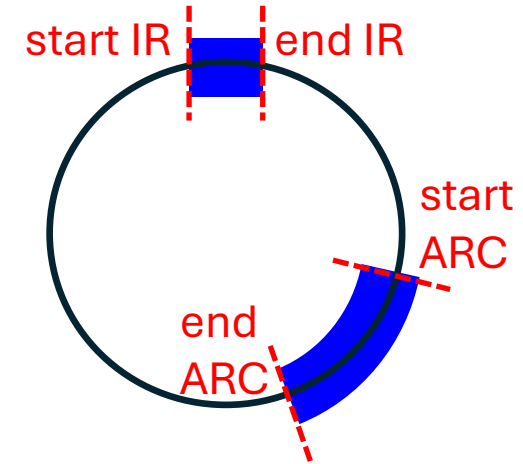


2. Components of the model

Model developed for FCC-ee, but not machine-specific → Can be extended to other machines

1. Grouping of elements

- Filtering via element type, name, or location
- Use of regex patterns or location markers



2. Components of the model

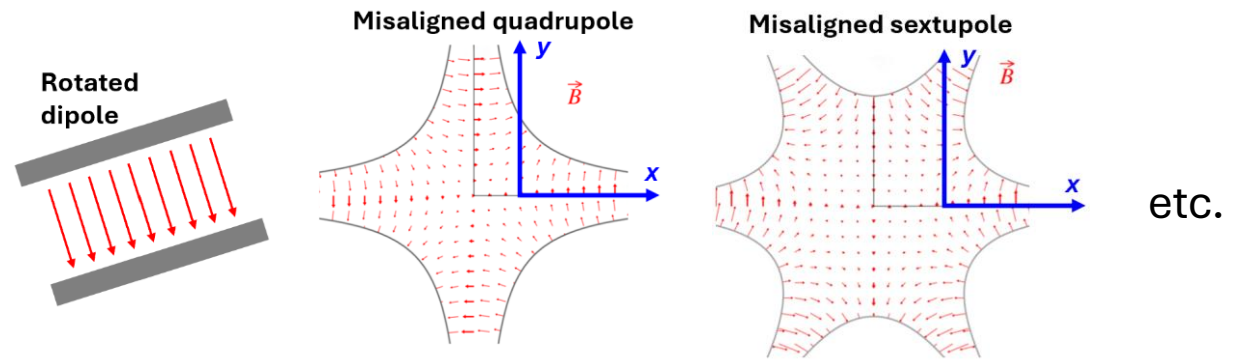
Model developed for FCC-ee, but not machine-specific → Can be extended to other machines

1. Grouping of elements

- Filtering via element type, name, or location
- Use of regex patterns or location markers

2. Error Assignment

- Applying randomised values to element attributes (misalignments, rotations, field errors, ...) for a specific element group
- Switches controlling the application of errors for a specific element group



Adapted from J. Wenninger, "[Linear Imperfections](#)", CAS 2020.

2. Components of the model

Model developed for FCC-ee, but not machine-specific → Can be extended to other machines

1. Grouping of elements

- Filtering via element type, name, or location
- Use of regex patterns or location markers

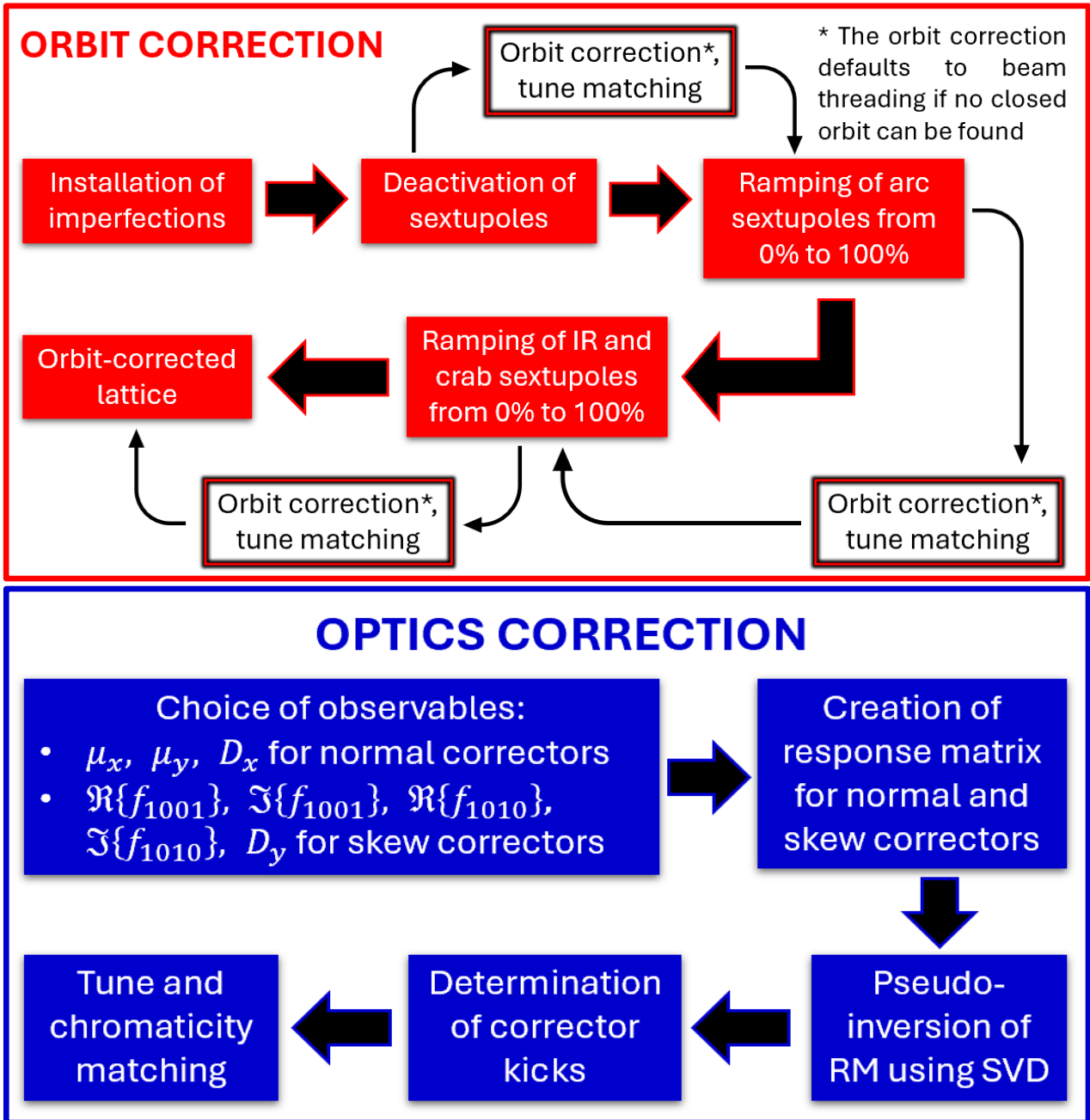
2. Error Assignment

- Applying randomised values to element attributes (misalignments, rotations, field errors, ...) for a specific element group
- Switches controlling the application of errors for a specific element group

3. Correction Machineries

- Installation of orbit and optics correctors
- Construction of response matrices
- Orbit and optics correction algorithms

Correction routine for FCC-ee



3. The model in practice (FCC-ee)

**Apply
imperfections**

```
# ARC QUADRUPOLES
apply_errors(line=line, seed=seed, sigmas=[50e-6, 50e-6, 100e-6, 50e-6], # <-- chosen error tolerances
            attrs=['shift_x', 'shift_y', 'shift_s', 'rot_s_rad_no_frame'], # <-- attributes to apply the errors to
            switch_name='on_misalignment_quad_arc', # <-- switch to turn on/off the errors
            element_names=arc_quad_names) # <-- lattice elements to apply the errors to
```

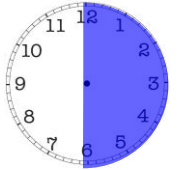


< 3min

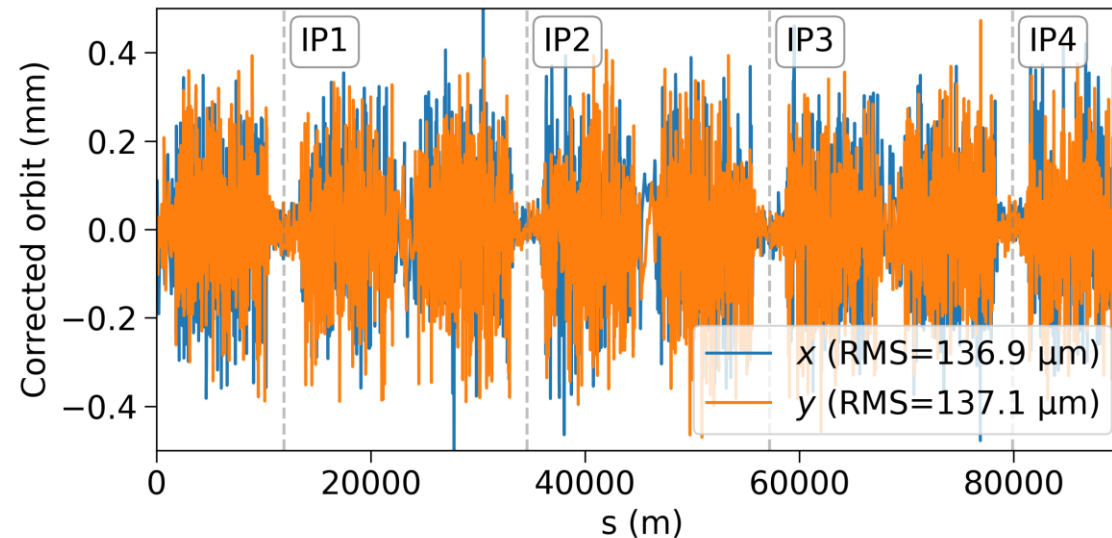
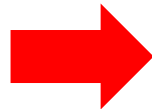
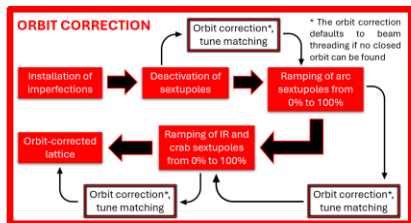
+ Prepare the lattice for correction (optimal BPM & corrector layout)

**Correct
orbit**

```
apply_orbit_correction(line, tw_ref, monitor_alignment, # <-- take the BPM alignment / read errors into account
                      num_sing_vals=num_sing_vals, # specify the number of singular values
                      corr_limit=corr_limit) # specify the corrector strength limits
```



15-45 min



3. The model in practice (FCC-ee)

Apply imperfections

```
# ARC QUADRUPOLES
apply_errors(line=line, seed=seed, sigmas=[50e-6, 50e-6, 100e-6, 50e-6], # <-- chosen error tolerances
            attrs=['shift_x', 'shift_y', 'shift_s', 'rot_s_rad_no_frame'], # <-- attributes to apply the errors to
            switch_name='on_misalignment_quad_arc', # <-- switch to turn on/off the errors
            element_names=arc_quad_names) # <-- lattice elements to apply the errors to
```

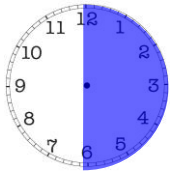


< 3min

+ Prepare the lattice for correction (optimal BPM & corrector layout)

Correct orbit

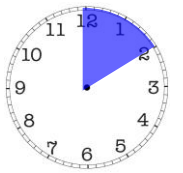
```
apply_orbit_correction(line, tw_ref, monitor_alignment, # <-- take the BPM alignment / read errors into account
                      num_sing_vals=num_sing_vals, # specify the number of singular values
                      corr_limit=corr_limit) # specify the corrector strength limits
```



15-45 min

Correct optics

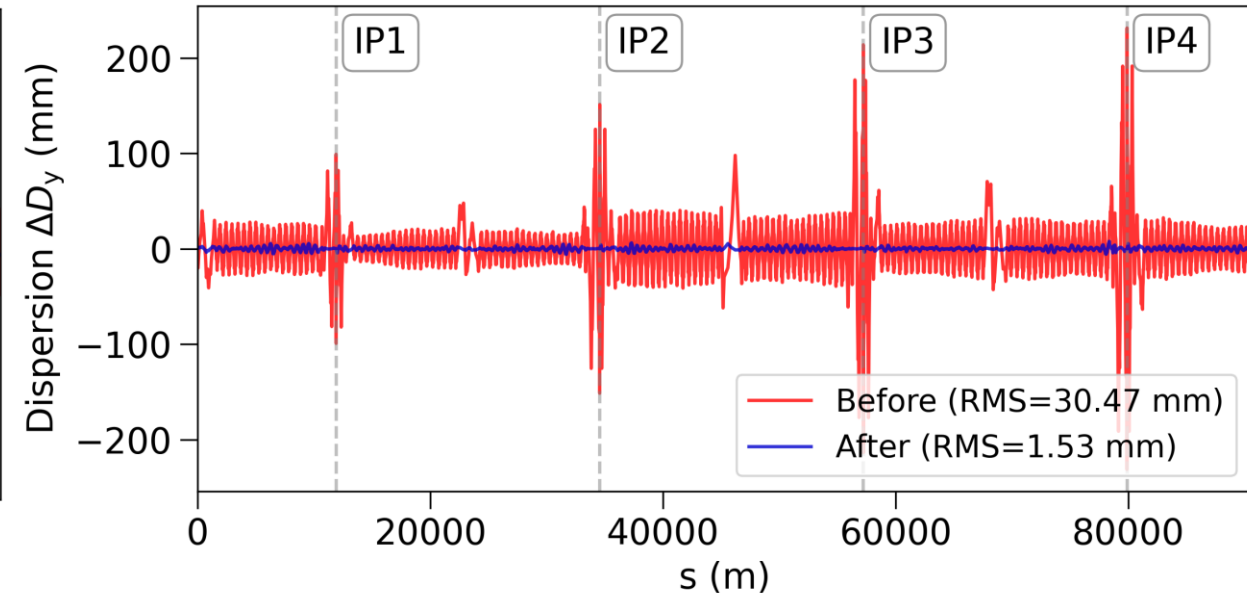
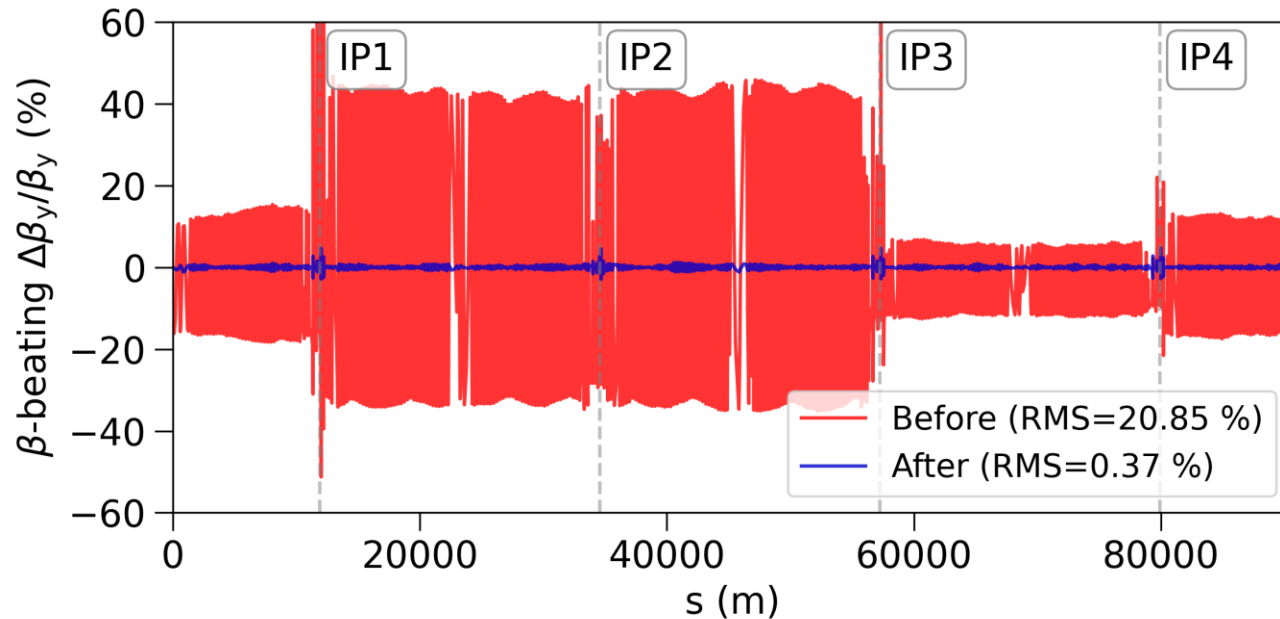
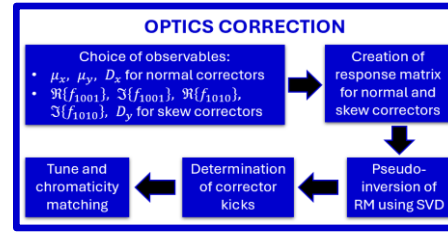
```
apply_optics_correction(line, RM_inv_normal, corr_knob_names_normal, # <-- inverse of normal RM and optics corrector names
                       tw_ref=tw_ref, OBSERVATION_POINTS=OBSERVATION_POINTS, # <-- reference Twiss and observation points
                       OBSERVABLES=OBSERVABLES_NORMAL, nloops=nloops) # <-- observables for normal RM and number of iterations
```



5-10 min

+ Response matrix generation (required only once): approx. 2 hours on HTCondor

3. The model in practice (FCC-ee)



**Correct
optics**

```
apply_optics_correction(line, RM_inv_normal, corr_knob_names_normal, # <-- inverse of normal RM and optics corrector names
                        tw_ref=tw_ref, OBSERVATION_POINTS=OBSERVATION_POINTS, # <-- reference Twiss and observation points
                        OBSERVABLES=OBSERVABLES_NORMAL, nloops=nloops) # <-- observables for normal RM and number of iterations
```



5-10 min

+ Response matrix generation (required only once): approx. 2 hours on HTCondor

II. Corrected FCC-ee Lattices (LCC Optics)

1. Assumed Error Tolerances

Section	Element	Misalignment			Rotation	Field error (rel.)
		$\sigma_x (\mu\text{m})$	$\sigma_y (\mu\text{m})$	$\sigma_s (\mu\text{m})$	$\sigma_\theta (\mu\text{rad})$	
ARC	Dipole	1000	1000	500	1000	$k_0: 1 \times 10^{-3}$
	Quadrupole	50	50	100	50	$k_1: 2 \times 10^{-4}$
	Sextupole	50	50	100	50	$k_2: 2 \times 10^{-4}$
	Girder	150	150	500	150	–
STRAIGHT SECTION	IR Dipole	1000	1000	100	1000	$k_0: 1 \times 10^{-3}$
	TR Dipole	1000	1000	500	1000	$k_0: 1 \times 10^{-3}$
	FD Quadrupole	30	30	100	30	$k_1: 5 \times 10^{-5}$
	FF Quadrupole (excl. FD)	30	30	100	30	$k_1: 1 \times 10^{-4}$
	TR Quadrupole	100	100	100	100	$k_1: 2 \times 10^{-4}$
	FF Sextupole (incl. CS)	30	30	100	30	$k_2: 1 \times 10^{-4}$
I	BPM	10	10	–	100	–

IR = Interaction Region

TR = Technical Region

FD = Final Doublet

FF = Final Focussing

CS = Crab Sextupole

1. Assumed Error Tolerances

Section	Element	Misalignment			Rotation	Field error (rel.)
		σ_x (μm)	σ_y (μm)	σ_s (μm)	σ_θ (μrad)	
ARC	Dipole	1000	1000	500	1000	$k_0: 1 \times 10^{-3}$
	Quadrupole	50	50	100	50	$k_1: 2 \times 10^{-4}$
	Sextupole	50	50	100	50	$k_2: 2 \times 10^{-4}$
	Girder	150	150	500	150	–
STRAIGHT SECTION	IR Dipole	1000	1000	100	1000	$k_0: 1 \times 10^{-3}$
	TR Dipole	1000	1000	500	1000	$k_0: 1 \times 10^{-3}$
	FD Quadrupole	30	30	100	30	$k_1: 5 \times 10^{-5}$
	FF Quadrupole (excl. FD)	30	30	100	30	$k_1: 1 \times 10^{-4}$
	TR Quadrupole	100	100	100	100	$k_1: 2 \times 10^{-4}$
	FF Sextupole (incl. CS)	30	30	100	30	$k_2: 1 \times 10^{-4}$
–	BPM	10	10	–	100	–

IR = Interaction Region

TR = Technical Region

FD = Final Doublet

FF = Final Focussing

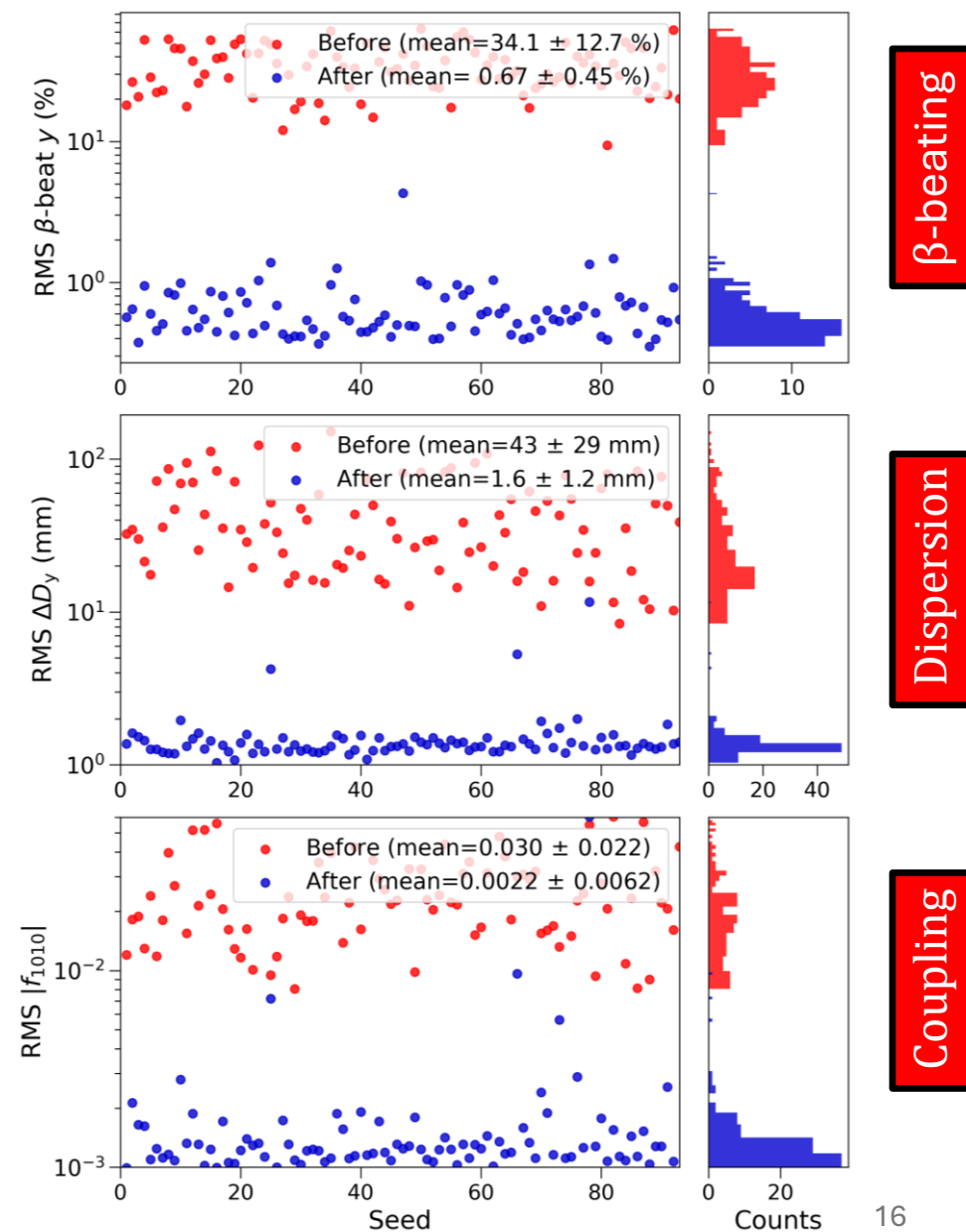
CS = Crab Sextupole

- Imperfections include:
 - **Misalignments** in transverse and longitudinal planes (x, y, s)
 - **Rotations** around the longitudinal axis (s -rotation)
 - **Field errors**
 - **BPM displacement errors**
- Imperfections are applied randomly based on a Gaussian distribution truncated at $2.5\sigma \rightarrow$ different “seeds”
- BPMs inherit imperfections from quadrupoles, which themselves may inherit from girder
- **Not yet included:** Higher-order errors, systematic errors, measurement errors

2. Tuning Results

Observable	LCC Z-pole (V106.2.3)
failed seeds	7 / 100
rms orbit H / V (μm)	137 / 138
rms H / V orbit corrector strength in ARC (μrad)	11.55 / 11.63
rms H / V orbit corrector strength in IR (μrad)	2.23 / 2.11
rms H / V orbit corrector strength in TR (μrad)	4.66 / 4.67
rms normal / skew optics corrector strength (mT)	0.77 / 0.50
rms $\Delta D_{x/y}$ (mm)	0.66 / 1.55
rms $\Delta\mu_{x/y}$ [10^{-4} 2π]	13.2 / 27.3
rms $\Delta\beta_{x/y}/\beta_{x/y}$ (%)	0.55 / 0.67
rms $ f_{1001} / f_{1010} $ [10^{-4}]	26.4 / 21.8
median ε_x (nm) / ε_y (pm) [from Twiss method]	0.71 / 0.028
transverse tunes Q_x/Q_y	194.16 / 170.20
rms $\Delta D_{x/y}^{(2)}$ (mm)	363 / 635

For corresponding table of K. André, see [optics tuning meeting 31/07/2026](#) (which includes all up-to-date error tolerances)



3. Tuned Lattices

All the lattices with corrected imperfections can be found on the following Github:

https://github.com/Sievert-L/FCCee_Project_2026


FCCee_Project_2026
Public
Pin
Watch 0
Fork 0
Star 0

main
1 Branch
0 Tags
Go to file
Add file
Code


Lara Sophie Sievert
Renamed folder
9dd52fb · 1 minute ago
79 Commits

	FCCee_LCC_Z_V105_imperfections_model	Renamed folder	1 minute ago
	FCCee_LCC_Z_V106_imperfections_model	Renamed folder	1 minute ago
	LEP3_Z_V11f_imperfections_model	Renamed folder	1 minute ago
	.gitattributes	Added RM helper files	last month
	README.md	Updated read.me	3 days ago
	helpers_for_imperfections_model.py	Uploads 106 version	3 days ago
	helpers_from_xutil.py	Uploads 106 version	3 days ago

README



Imperfections Model of the FCC-ee with Corrections and Beam-Beam Effects

Master's Project 2026

Lara Sophie Sievert (Supervisors: Dr. Tirsi Prebibaj, Dr. Tatiana Pieloni, Prof. Dr. Mike Seidel)

About

Master's Project on the development of an imperfections model of the FCC-ee with corrections and beam-beam effects

 Readme

 Activity

 0 stars

 0 watching

 0 forks

Releases

No releases published

[Create a new release](#)

Packages

No packages published

[Publish your first package](#)

Contributors 1


Sievert-L Lara Sophie Sievert

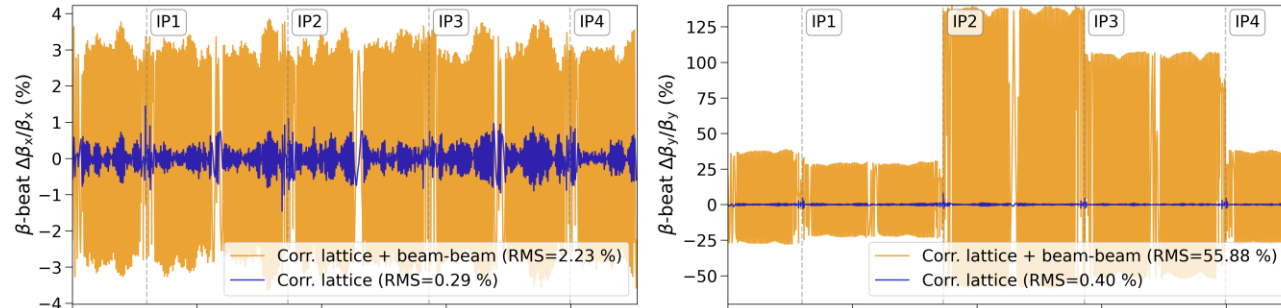
Languages

III. Beam-Beam and Other Studies with these Tuned Lattices

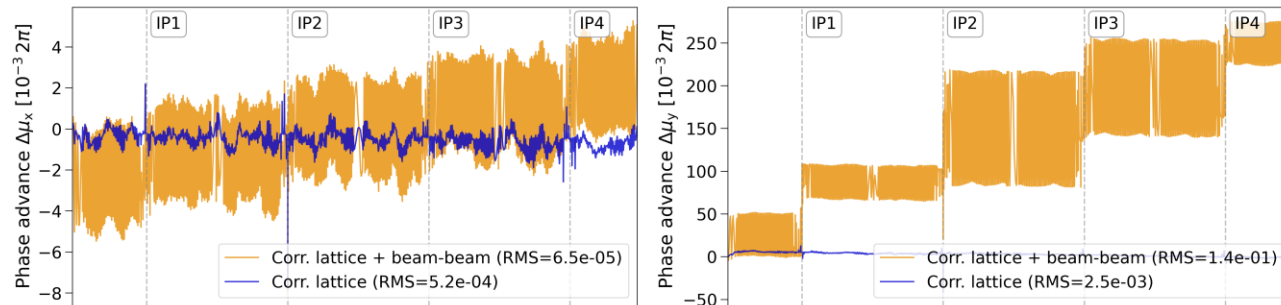
1. Beam-Beam and Impact on Optics

- Quantifying beam-beam induced **optics degradation** in lattices with imperfections
→ e.g. see [IPAC paper](#)

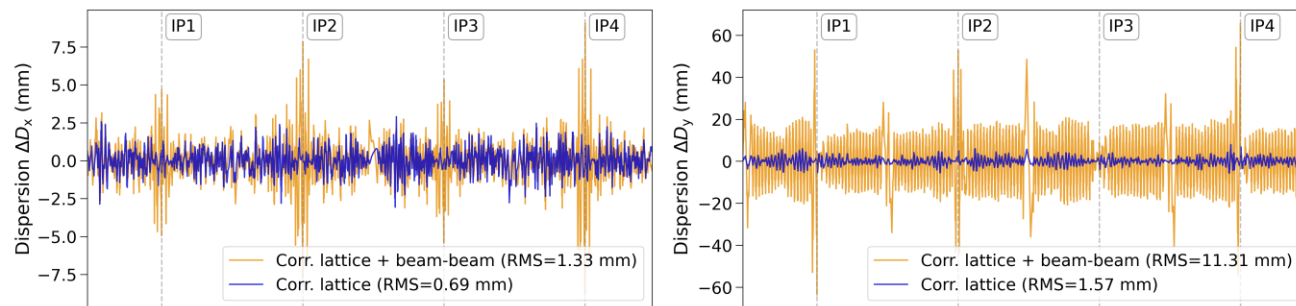
β -beating



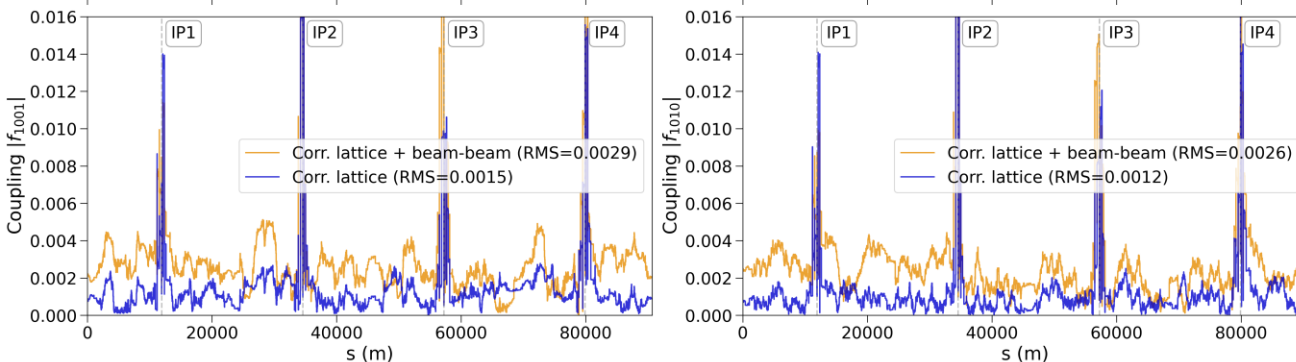
Phase adv.



Dispersion



Coupling



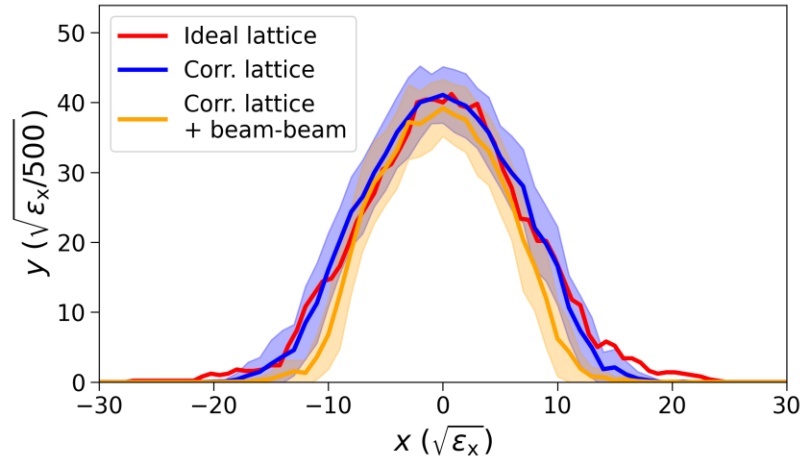
Beam-beam parameter

$$\xi_x = 0.0014 \text{ and } \xi_y = 0.09$$

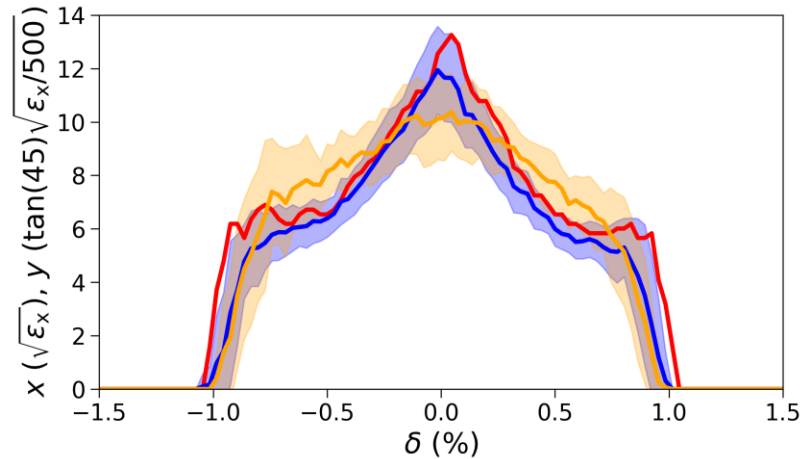
2. Beam-Beam and Impact on Performance

Dynamic Aperture / Momentum Acceptance

DA



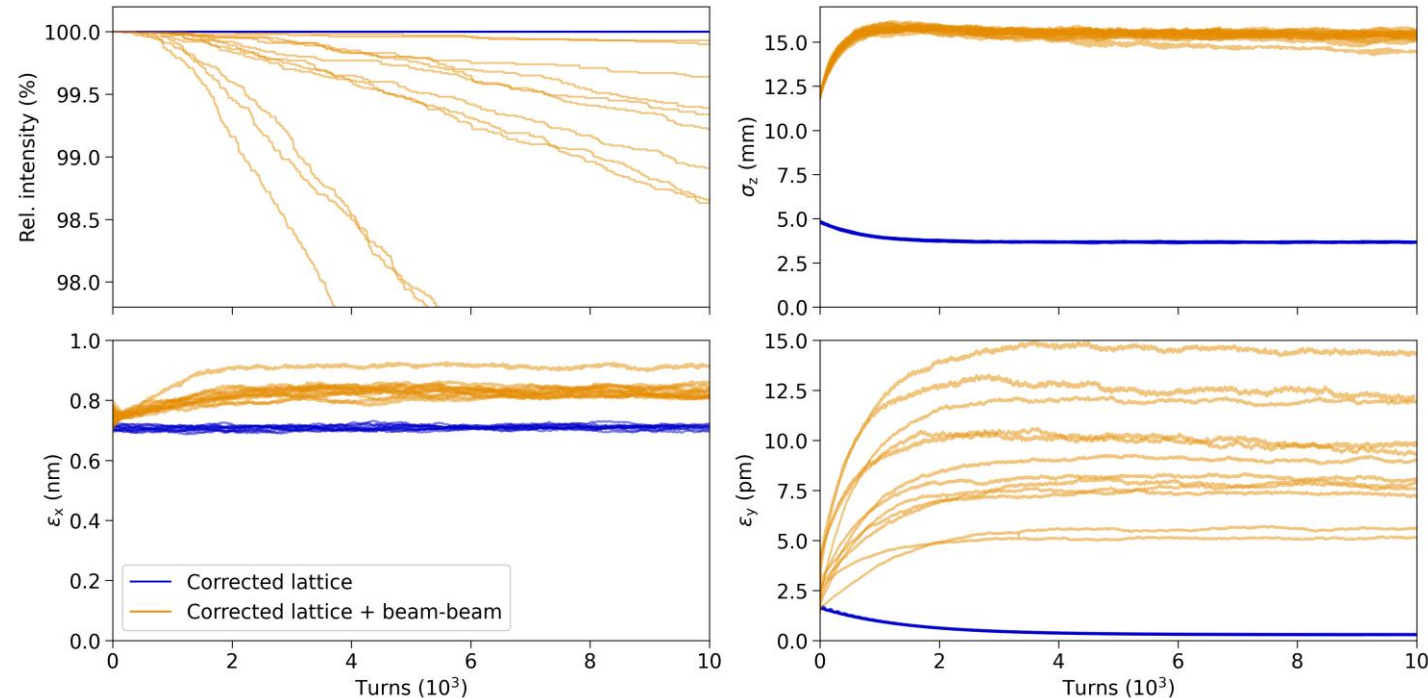
MA



- Quantifying beam-beam induced **optics degradation** in lattices with imperfections
→ e.g. see [IPAC paper](#)
- Quantifying beam-beam induced luminosity / **performance degradation** in lattices with imperfections
→ e.g. see [Master's thesis oral defence](#)

2. Beam-Beam and Impact on Performance

Tracking results



- Quantifying beam-beam induced **optics degradation** in lattices with imperfections
→ e.g. see [IPAC paper](#)
- Quantifying beam-beam induced luminosity / **performance degradation** in lattices with imperfections
→ e.g. see [Master's thesis oral defence](#)

Details on simulation setup

- Tracking 10⁴ particles for 10⁴ turns in the corrected lattices
- Artificial aperture at 20 σ in both transverse planes, RF bucket length as the limit in the longitudinal plane
- Synchrotron radiation with quantum excitations, Beamstrahlung, and tapering are included
- Beam-beam interactions are modelled using the “quasi-strong-strong” scheme

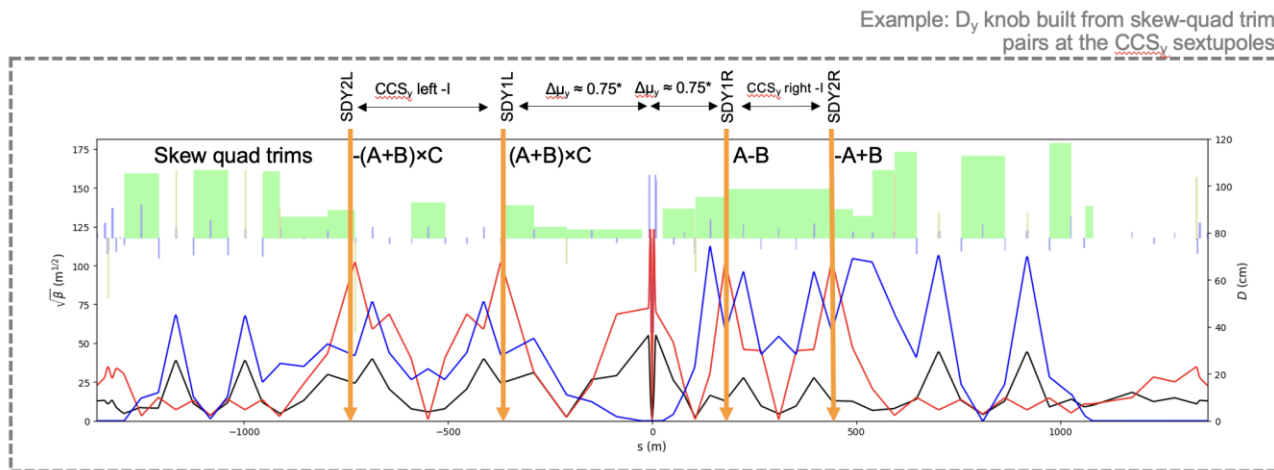
3. Tuning Knobs

- Knobs for correcting residuals at IPs with or without collisions
→ e.g. *IP dispersion, waist, phase, coupling*

Developed by **Tirsi**

Example: **Dispersion knob**

(see [beam-beam meeting Tirsi 26/05/2026](#))

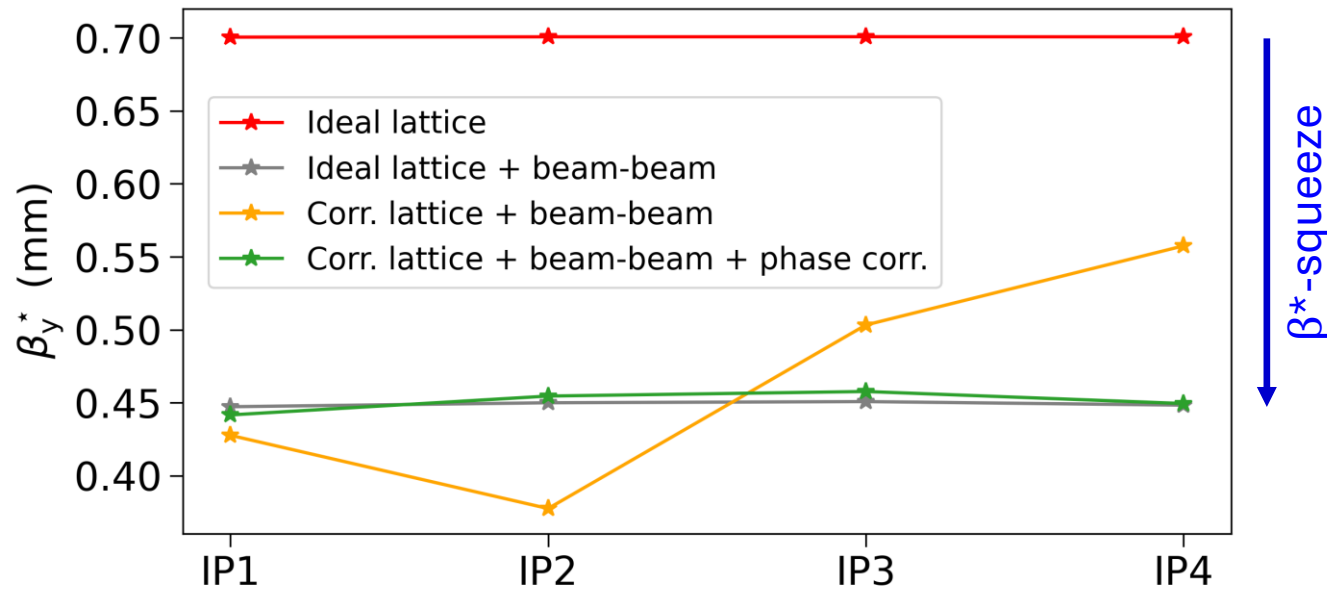


- Quantifying beam-beam induced **optics degradation** in lattices with imperfections
→ e.g. see [IPAC paper](#)
- Quantifying beam-beam induced luminosity / **performance degradation** in lattices with imperfections
→ e.g. see [Master's thesis oral defence](#)
- Developing IP **knobs** for recovering optics and performance with beam-beam
→ e.g. see [FCC week presentation of Tirsi & beam-beam meeting 26/05/2026](#)

3. Tuning Knobs

- Knobs for correcting residuals at IPs with or without collisions
 - e.g. *IP dispersion, waist, phase, coupling*
- Evaluating impact of IP tuning + beam-beam in tuned lattices for many seeds
 - *On the optics*

Example: **Restoring β^* symmetry between IPs**
(see [Master's thesis oral defence](#))

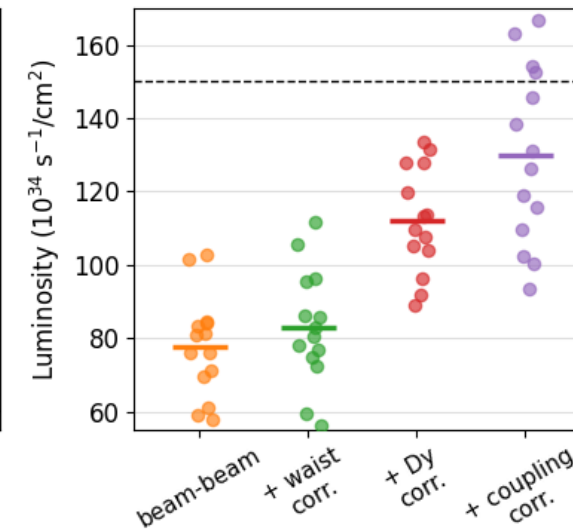
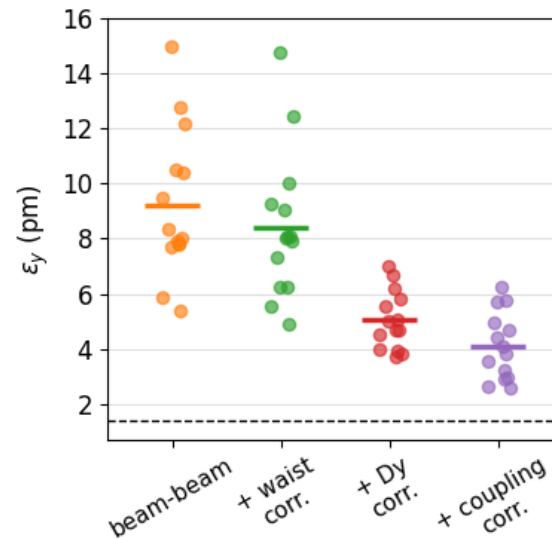
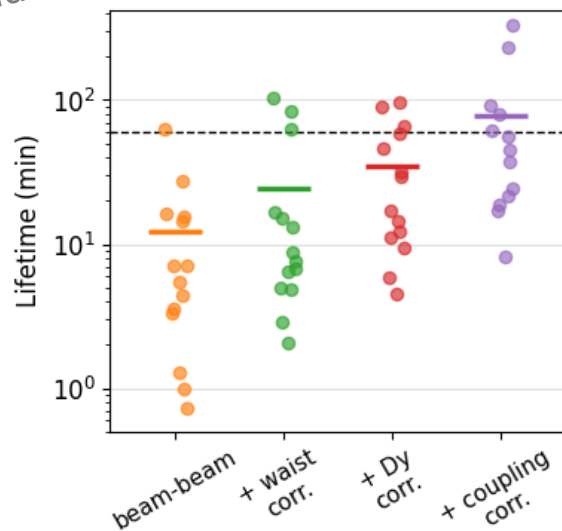


- Quantifying beam-beam induced **optics degradation** in lattices with imperfections
 - e.g. see [IPAC paper](#)
- Quantifying beam-beam induced luminosity / **performance degradation** in lattices with imperfections
 - e.g. see [Master's thesis oral defence](#)
- Developing IP **knobs** for recovering optics and performance with beam-beam
 - e.g. see [FCC week presentation of Tirsi & beam-beam meeting 26/05/2026](#)

3. Tuning Knobs

- Knobs for correcting residuals at IPs with or without collisions
 - e.g. *IP dispersion, waist, phase, coupling*
- Evaluating impact of IP tuning + beam-beam in tuned lattices for many seeds
 - On the optics
 - On the performance (lifetime, emittance, luminosity)

In collaboration with the tuning WG, the knobs are used/tested in other imperfection models (e.g. K. André in pyAT)

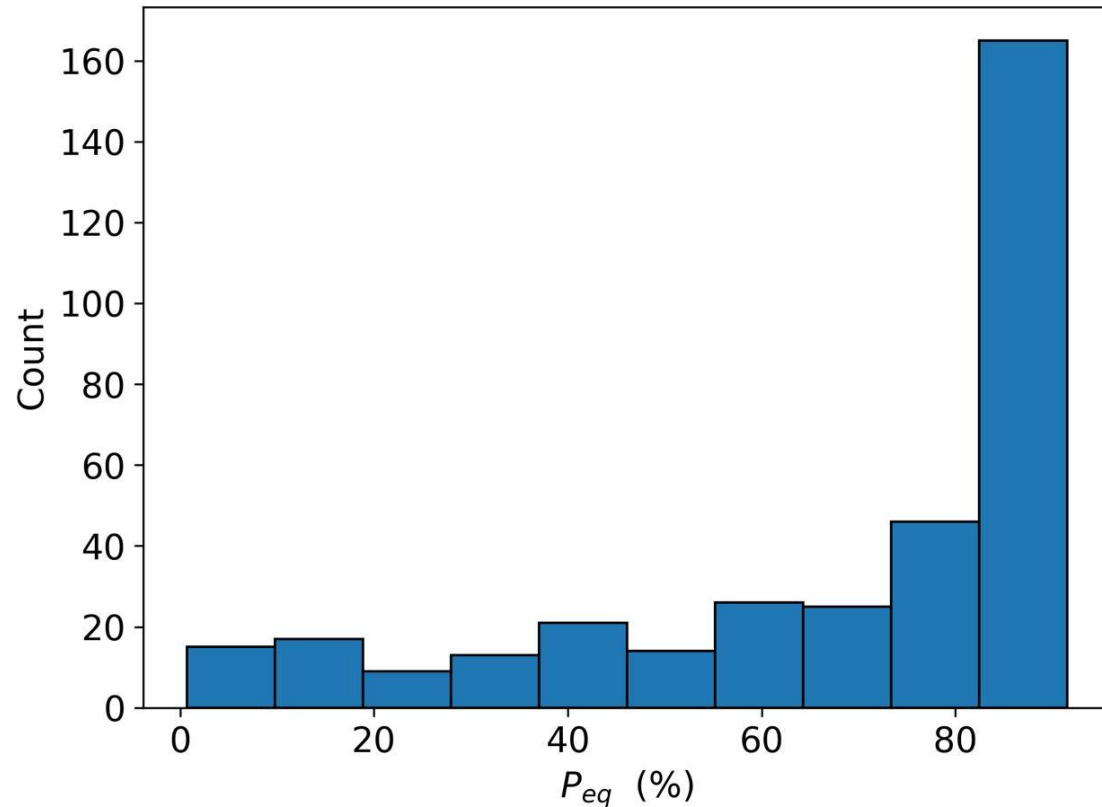


Presented by **Tirsi in [beam-beam meeting](#) and [FCC week](#)**

- Quantifying beam-beam induced **optics degradation** in lattices with imperfections
 - e.g. see [IPAC paper](#)
- Quantifying beam-beam induced luminosity / **performance degradation** in lattices with imperfections
 - e.g. see [Master's thesis oral defence](#)
- Developing IP **knobs** for recovering optics and performance with beam-beam
 - e.g. see [FCC week presentation of Tirsi & beam-beam meeting 26/05/2026](#)

4. Spin Polarisation Studies

Equilibrium polarisation
in tuned lattices

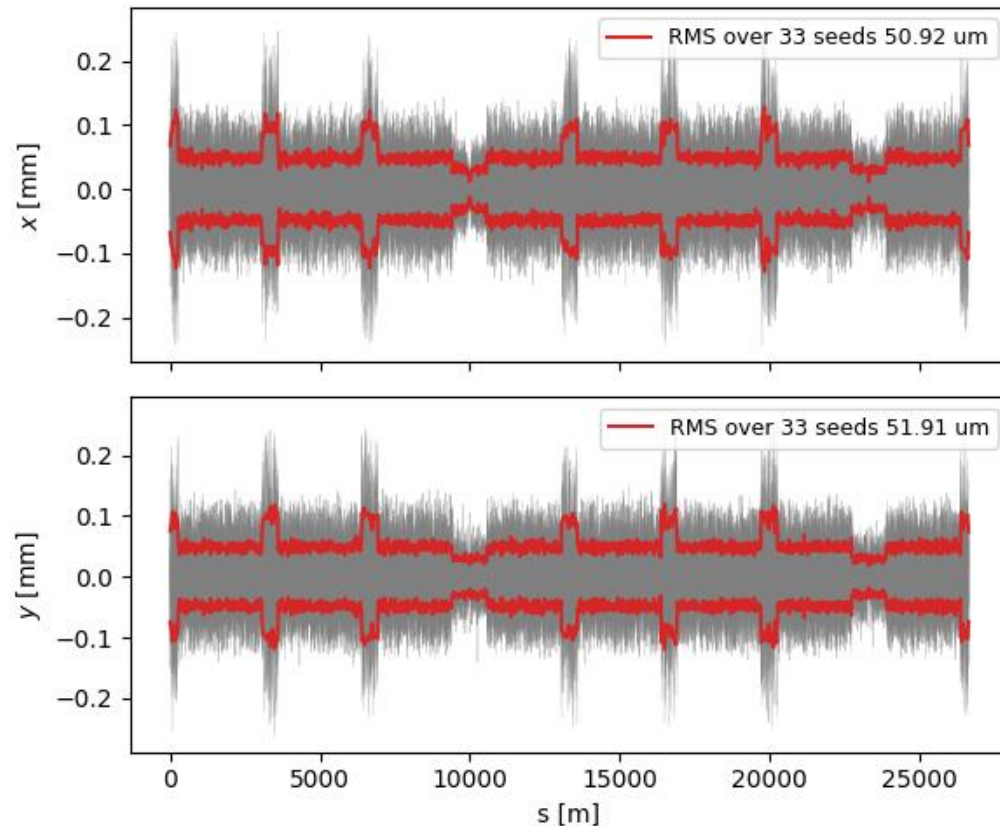


Yi Wu

- Quantifying beam-beam induced **optics degradation** in lattices with imperfections
→ e.g. see [IPAC paper](#)
- Quantifying beam-beam induced luminosity / **performance degradation** in lattices with imperfections
→ e.g. see [Master's thesis oral defence](#)
- Developing IP **knobs** for recovering optics and performance with beam-beam
→ e.g. see [FCC week presentation of Tirsi & beam-beam meeting 26/05/2026](#)
- Performing **polarisation** studies
→ e.g. see [FCC week presentation of Yi](#)

5. Imperfections Model for LEP3

Corrected orbit in LEP3 v11f @ Z



- Quantifying beam-beam induced **optics degradation** in lattices with imperfections
→ e.g. see [IPAC paper](#)
- Quantifying beam-beam induced luminosity / **performance degradation** in lattices with imperfections
→ e.g. see [Master's thesis oral defence](#)
- Developing IP **knobs** for recovering optics and performance with beam-beam
→ e.g. see [FCC week presentation of Tirsi & beam-beam meeting 26/05/2026](#)
- Performing **polarisation** studies
→ e.g. see [FCC week presentation of Yi](#)
- Extending **tuning to non-FCC lattices** (e.g. LEP3)
→ work in progress

Conclusions

- Developed Xsuite model for lattice imperfections with orbit and optics corrections to study beam-beam in tuned lattices
- Master's thesis focused on FCC-ee LCC lattice @ Z, but generic implementation of model allows for it to be extended to other machines
- Tuned lattices all available on Github: https://github.com/Sievert-L/FCCee_Project_2026
- Significant beam-beam induced optics degradations observed in tuned lattices, with a severe impact on collider performance
- Beam-beam tuning knobs to restore optics and recover collider performance
- Tuned lattices also used in spin polarisation studies, extending the model to LEP3 lattices seems straightforward

Thank you!

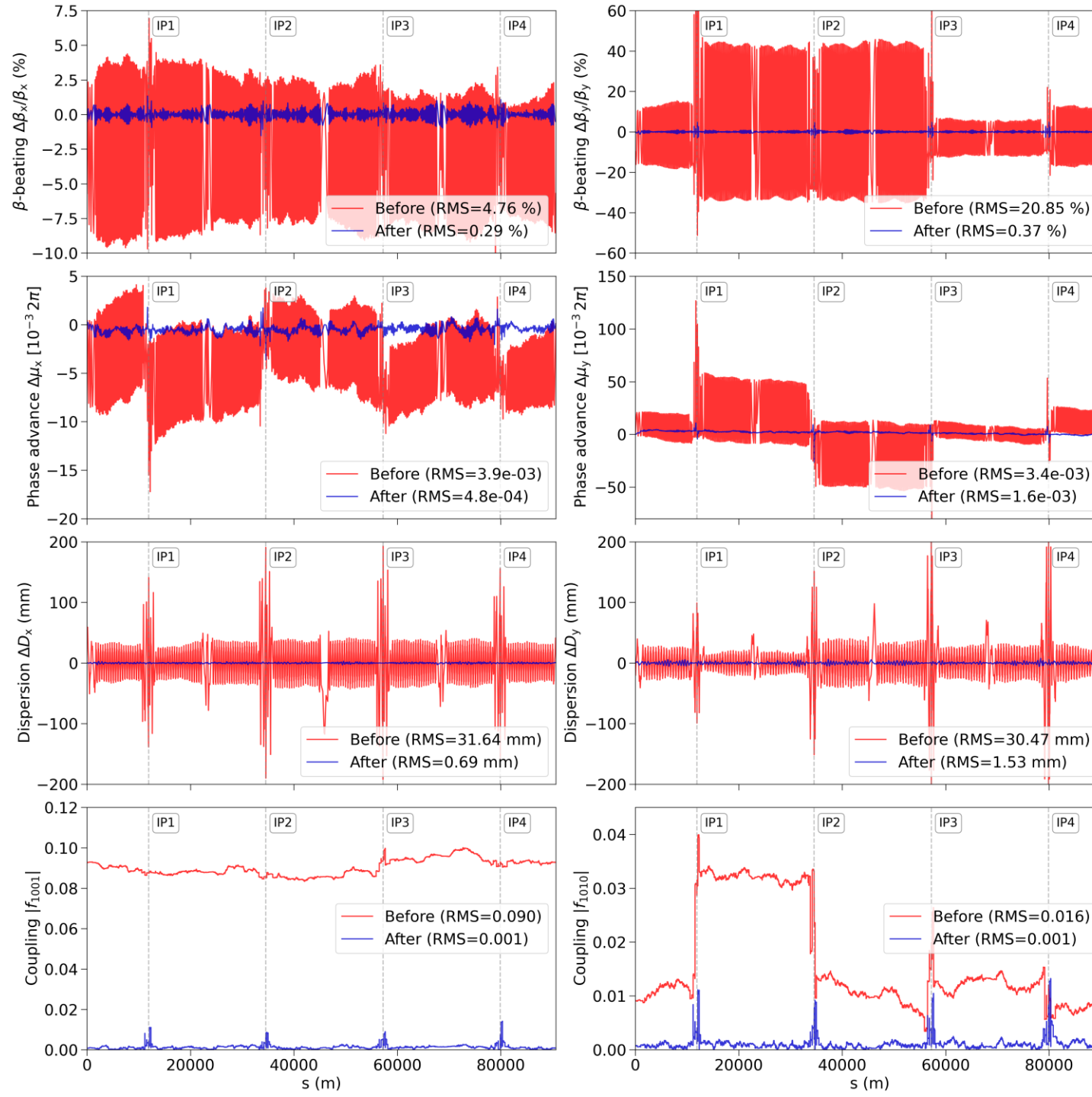
Back Up

Parameter Table for GHC & LCC Lattices

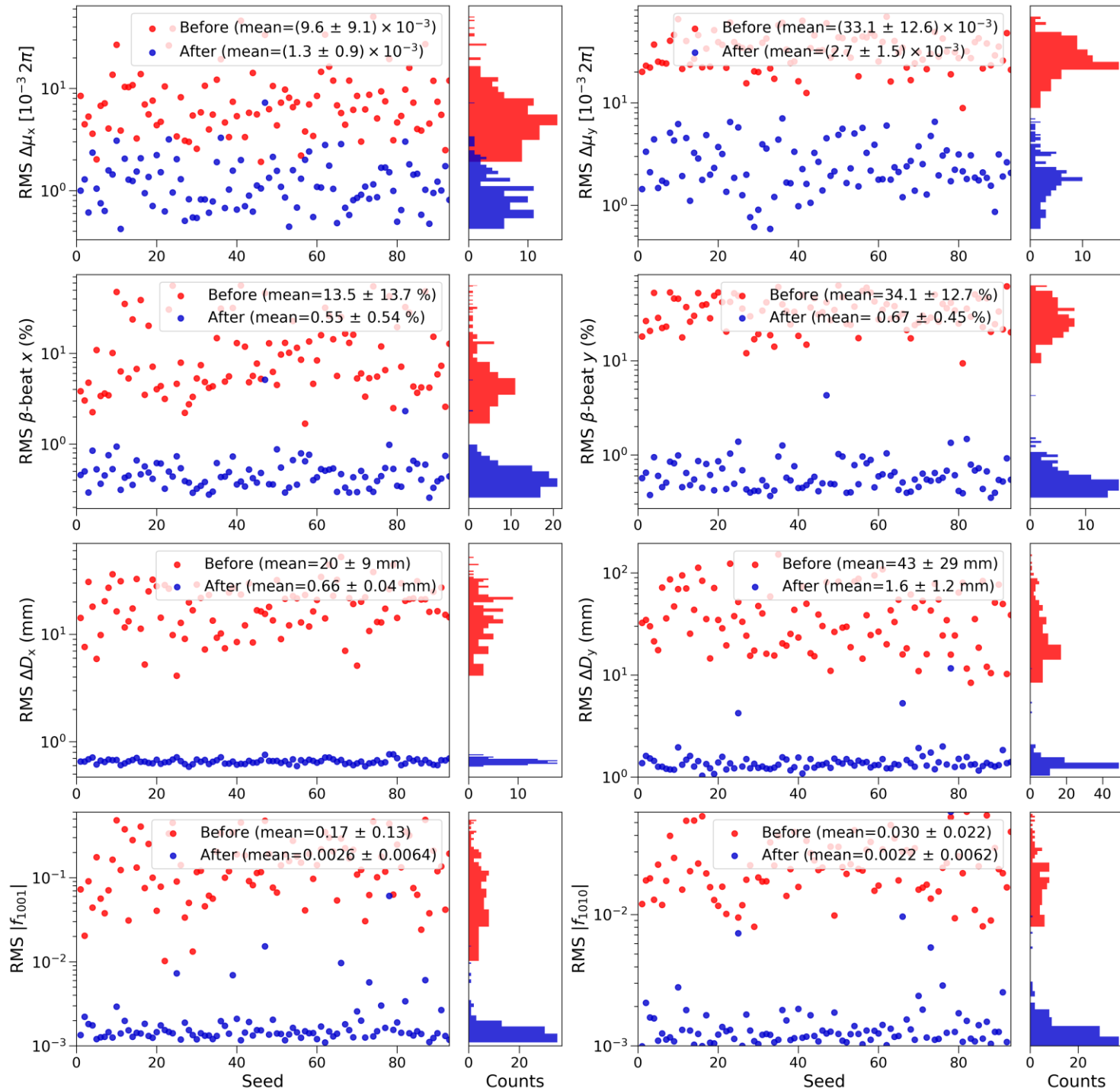
Optics Mode	GHC		LCC	
	Z	$\bar{t}\bar{t}$	Z	$\bar{t}\bar{t}$
Beam energy E (GeV)	45.6	182.5	45.6	182.5
Circumference C (m)	90658.525		90644.816	
Arc-cell setup	90°/90° long	90°/90° short	52°/45°	99°/77°
Momentum compaction factor α_c (10^{-6})	28.5	7.3	28.6	9.5
Energy Loss per turn W_0 (GeV)	0.039	10.01	0.035	9.01
Beam Intensity N (10^{12} particles)	2400	9.405	2424	9.435
Bunch Intensity N_b (10^{11} particles)	2.02	1.85	2.02	2.20
Number of Bunches	12000	51	12000	43
Horizontal β -function at IP β_x^* (cm)	9	90	9	90
Vertical β -function at IP β_y^* (mm)	0.7	14	0.7	14
Horizontal emittance ϵ_x (nm)	0.74	1.74	0.70	2.10
Target vertical emittance in collision ϵ_y (pm)	1.48	1.75	1.40	2.11
Transverse tune Q_x/Q_y	214.16 / 214.20	394.19 / 390.27	194.16 / 170.20	346.19 / 262.27
Chromaticity Q'_x/Q'_y	12 / 5	0 / 0	12 / 5	0 / 0
Harmonic number h at 400 MHz	121200		121200	
Total RF voltage of 400 / 800 MHz (GV)	0.09	2.1 / 8.9	0.09	2.0 / 8.1
Synchrotron tune Q_s	0.031	0.086	0.031	0.111
RF momentum acceptance (%)	1	2	1	2
Bunch length σ_z (non coll./coll.) (mm)	5.4 / 16.3	1.9 / 2.7	5.1 / 16.7	1.9 / 2.8
Rel. mom. spread σ_p (non coll./coll.) (10^{-3})	0.40 / 1.29	1.58 / 2.21	0.39 / 1.34	1.52 / 2.33
Longitudinal damping time τ_z (turns)	1159	18	1297	20
Crab-waist ratio (%)	55	40	55	40
Beam-beam parameter ξ_x/ξ_y (10^{-3})	1.5 / 80	61.2 / 108.9	1.4 / 80	67.1 / 120
Luminosity $\mathcal{L}$ ($10^{34} \text{ s}^{-1}/\text{cm}^2$)	151	1.54	150	1.45

Source: [Optics Comparison Report](#)

Tuning results for one seed



Tuning results across 100 seeds

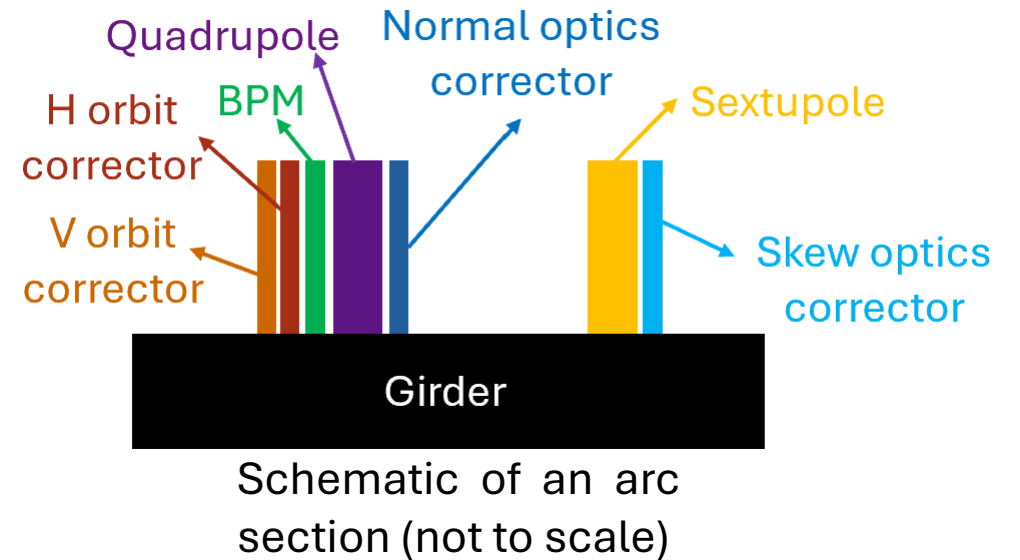


Results for tuned lattices with beam-beam @ 75 %

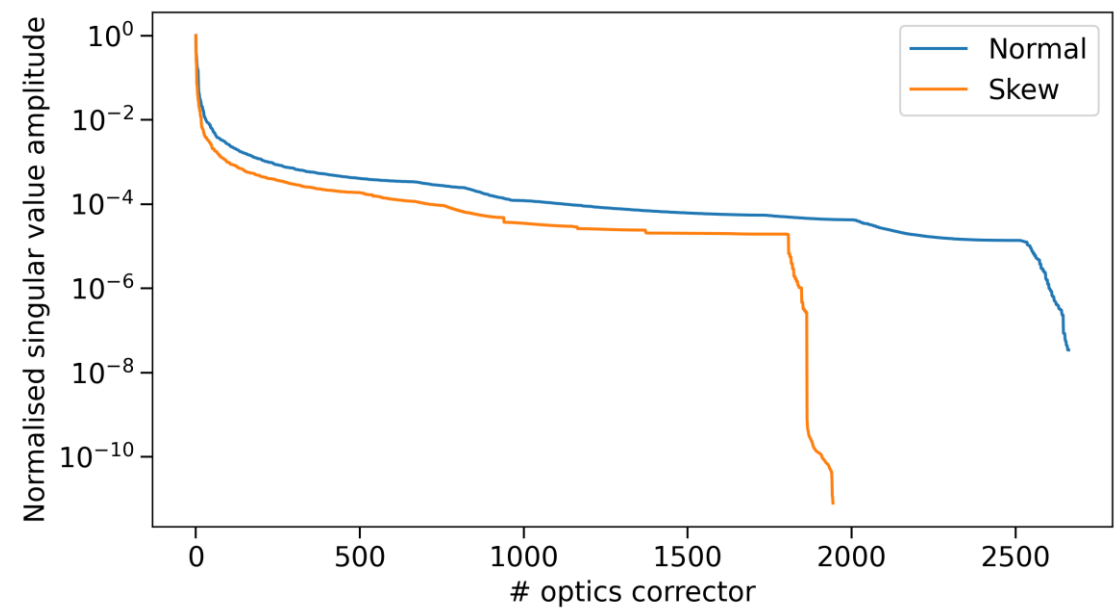
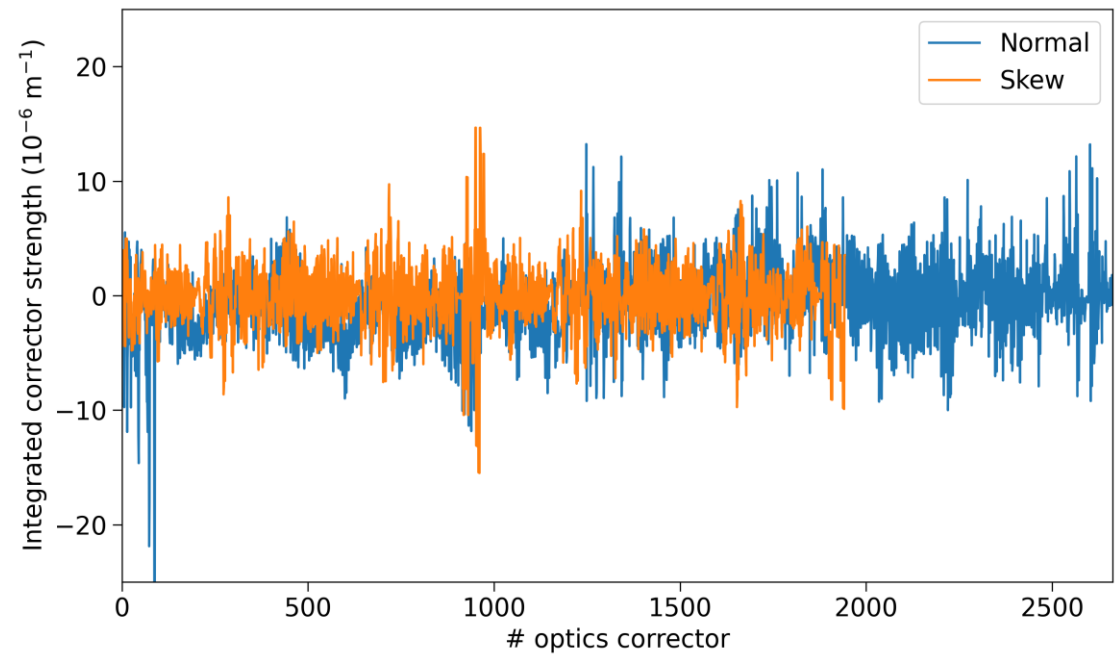
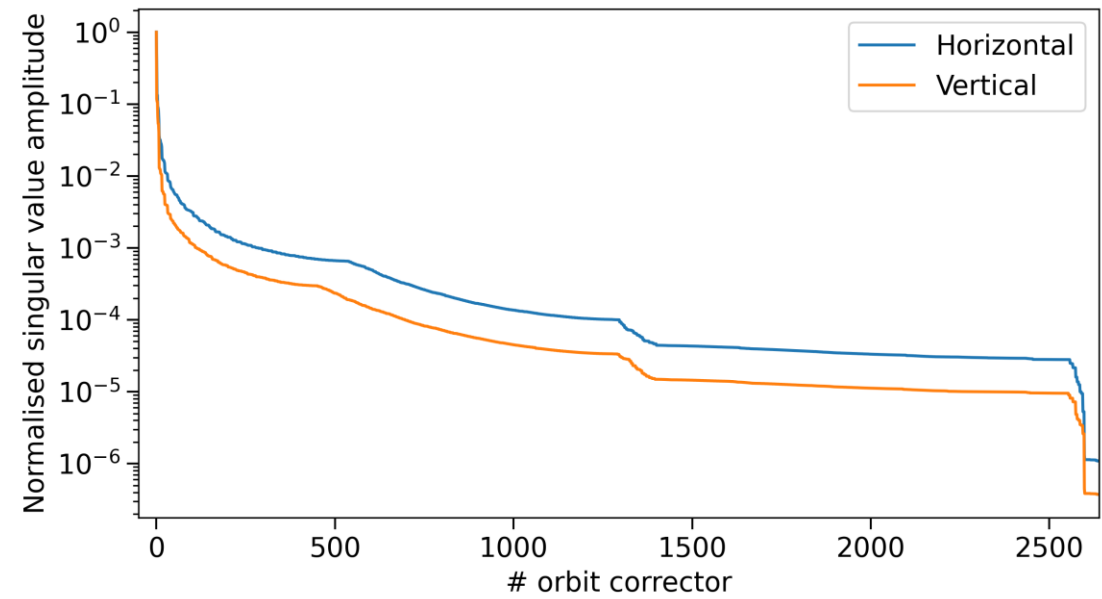
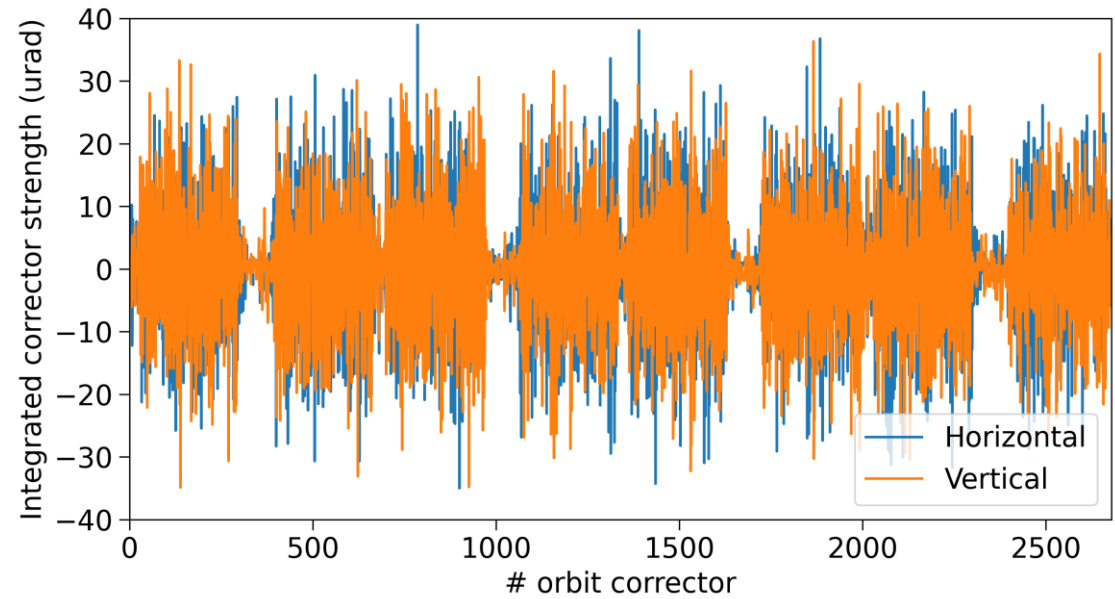
Observable	LCC Z-pole (V106.2.3)
number of lattices	82
rms $\Delta D_{x/y}$ (mm)	1.6 / 9.0
rms $\Delta\mu_{x/y}$ [10^{-4} 2π]	27.1 / 1216
rms $\Delta\beta_{x/y}/\beta_{x/y}$ (%)	2.12 / 42.7
rms $ f_{1001} / f_{1010} $ [10^{-4}]	49.3 / 41.5
median ε_x (nm) / ε_y (pm)	0.72 / 0.33
transverse tunes Q_x/Q_y	194.165 / 170.405
rms $\Delta D_{x/y}^{(2)}$ (mm)	1049 / 1563

Details on lattice layout

Element type	# Elements
Dipole	2432
Quadrupole	2679
Sextupole	1952
BPM	2639
H / V orbit corrector	2679
Normal optics corrector	2663
Skew optics corrector	1944



Details on orbit / optics correction



Overview of Imperfections

	Field type	Imperfection	Error type	Impact
a)	Dipole	Field error	Dipole	Orbit, trajectory, energy
b)	Dipole	Roll	Dipole	Orbit, trajectory
	Quadrupole	Field error	Quadrupole	Tune, optics
c)	Quadrupole	Offset	Dipole	Orbit, trajectory
	Quadrupole	Roll	Skew quadrupole	Coupling
	Sextupole	Field error	Sextupole	Chromaticity
d)	Sextupole	Horizontal offset	Quadrupole	Tune, optics
	Sextupole	Vertical offset	Skew quadrupole	Coupling

J. Wenninger, “[Linear Imperfections](#)”, CAS 2020.

Orbit Distortion

$$u(s) = \frac{\sqrt{\beta(s)}}{2 \sin(\pi Q)} \oint \theta(\bar{s}) \sqrt{\beta(\bar{s})} \cos(\pi Q - |\mu(s) - \mu(\bar{s})|) d\bar{s}$$

Kicks θ at locations $\bar{s}$

Optics Distortion

• Tune shift

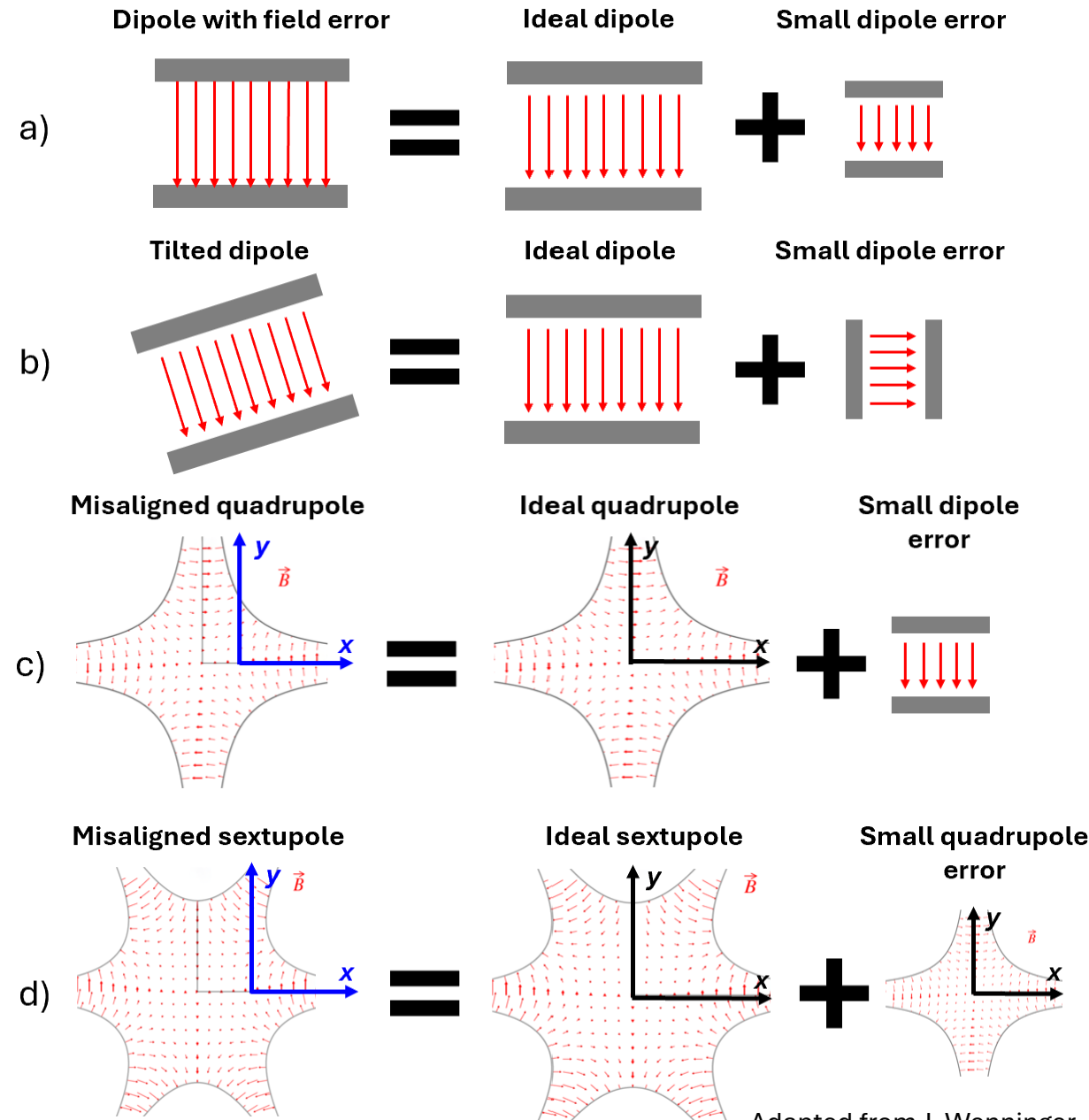
$$\Delta Q = \frac{1}{4\pi} \oint \delta K(\bar{s}) \beta(\bar{s}) d\bar{s}$$

u – Transverse coordinate (x or y)
 s – Longitudinal coordinate
 $\bar{s}$ – Location of the error

• Beta-beating

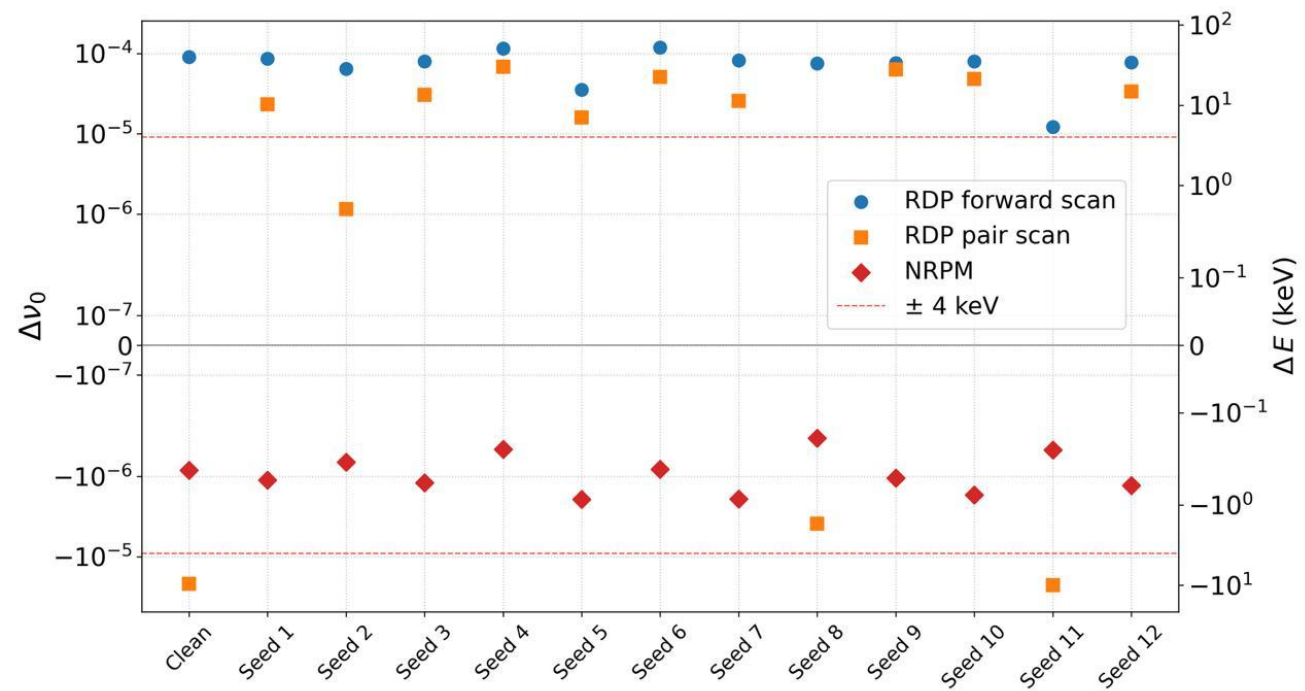
$$\frac{\Delta\beta(s)}{\beta(s)} = \frac{1}{2 \sin(2\pi Q)} \oint \beta(\bar{s}) \delta K(\bar{s}) \cos(2|\mu(s) - \mu(\bar{s})| - 2\pi Q) d\bar{s}$$

Field errors δK at locations $\bar{s}$



Adapted from J. Wenninger.

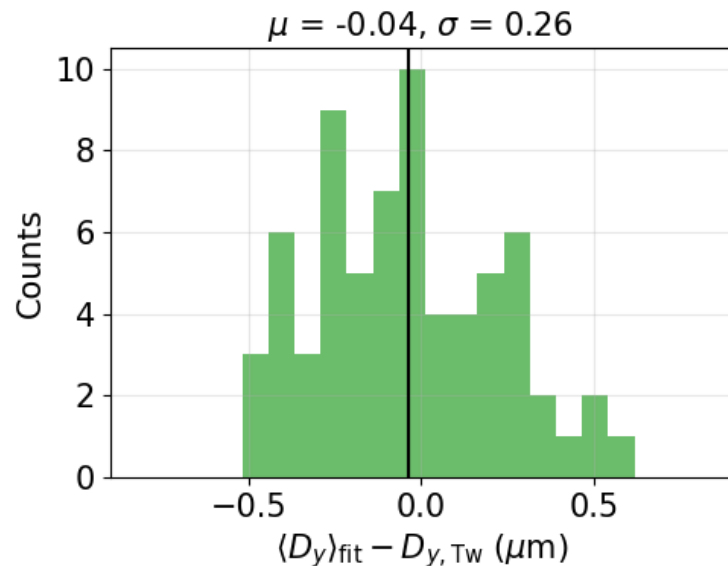
Spin Polarisation Studies



Yi Wu

5. Dispersion Measurement Studies

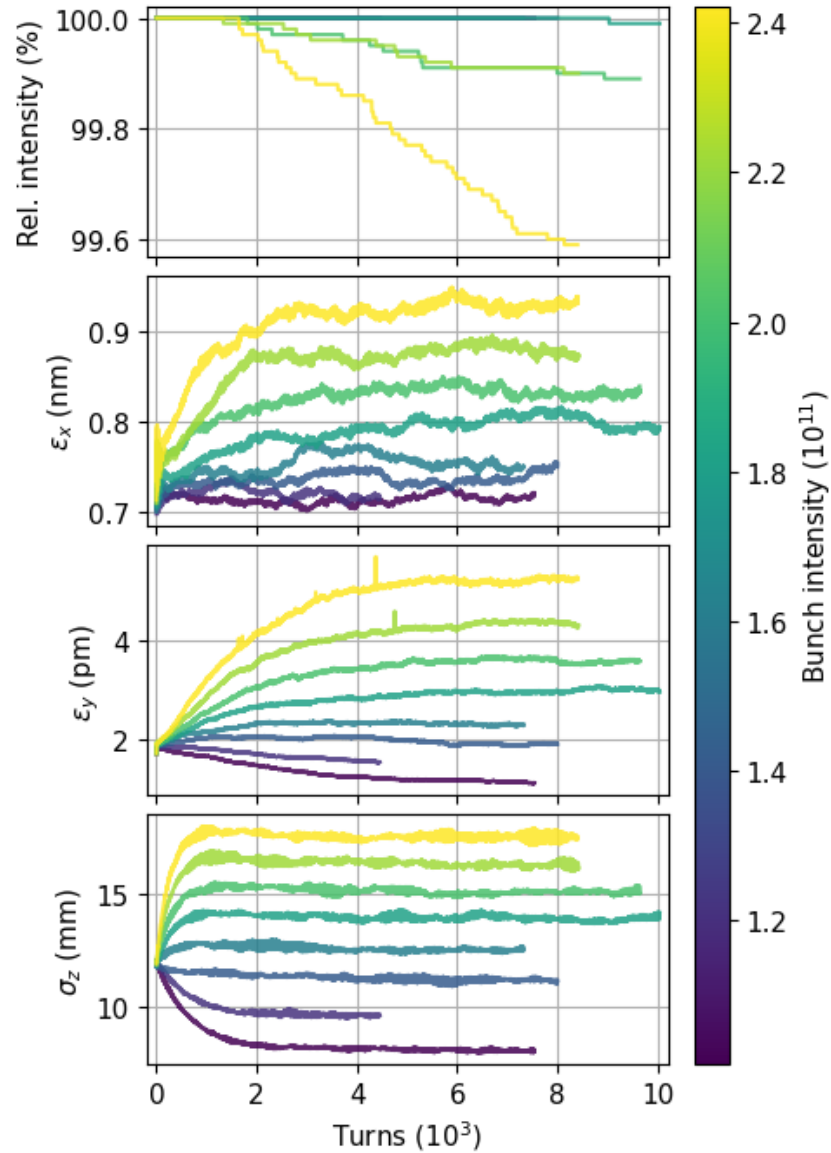
Vertical dispersion reconstruction via radial steering in tuned lattices (multiple seeds)



Tirsi Prebibaj

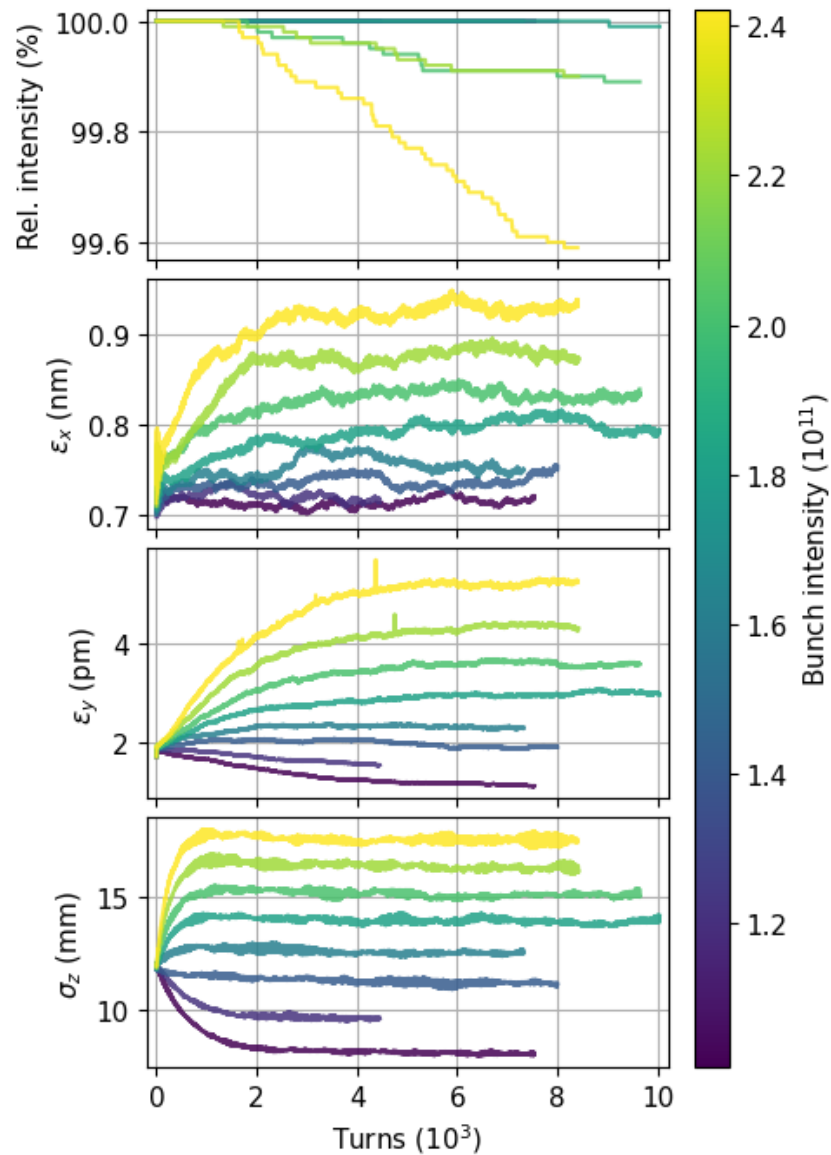
- Quantifying beam-beam induced **optics degradation** in lattices with imperfections
→ e.g. see [IPAC paper](#)
- Quantifying beam-beam induced luminosity / **performance degradation** in lattices with imperfections
→ e.g. see [Master's thesis oral defence](#)
- Developing IP **knobs** for recovering optics and performance with beam-beam
→ e.g. see [FCC week presentation of Tirsi & beam-beam meeting 26/05/2026](#)
- Performing **polarisation** studies
→ e.g. see [FCC week presentation of Yi](#)
- Performing **dispersion measurement** studies
→ e.g. see [beam-beam meeting 18/08/2026](#)

Intensity scan



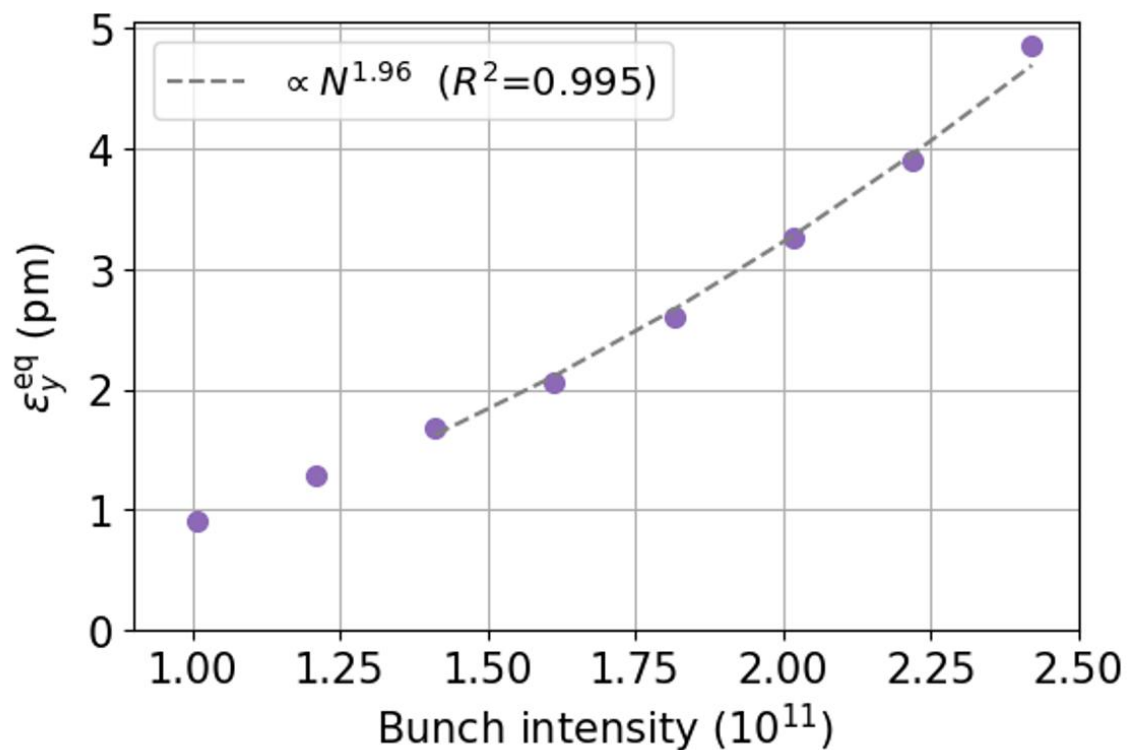
For a single seed with the tuning knobs + beam-beam, bunch intensity was scanned ($1 \times 10^{11} \rightarrow 2.4 \times 10^{11}$).

Probing the beam-beam limit

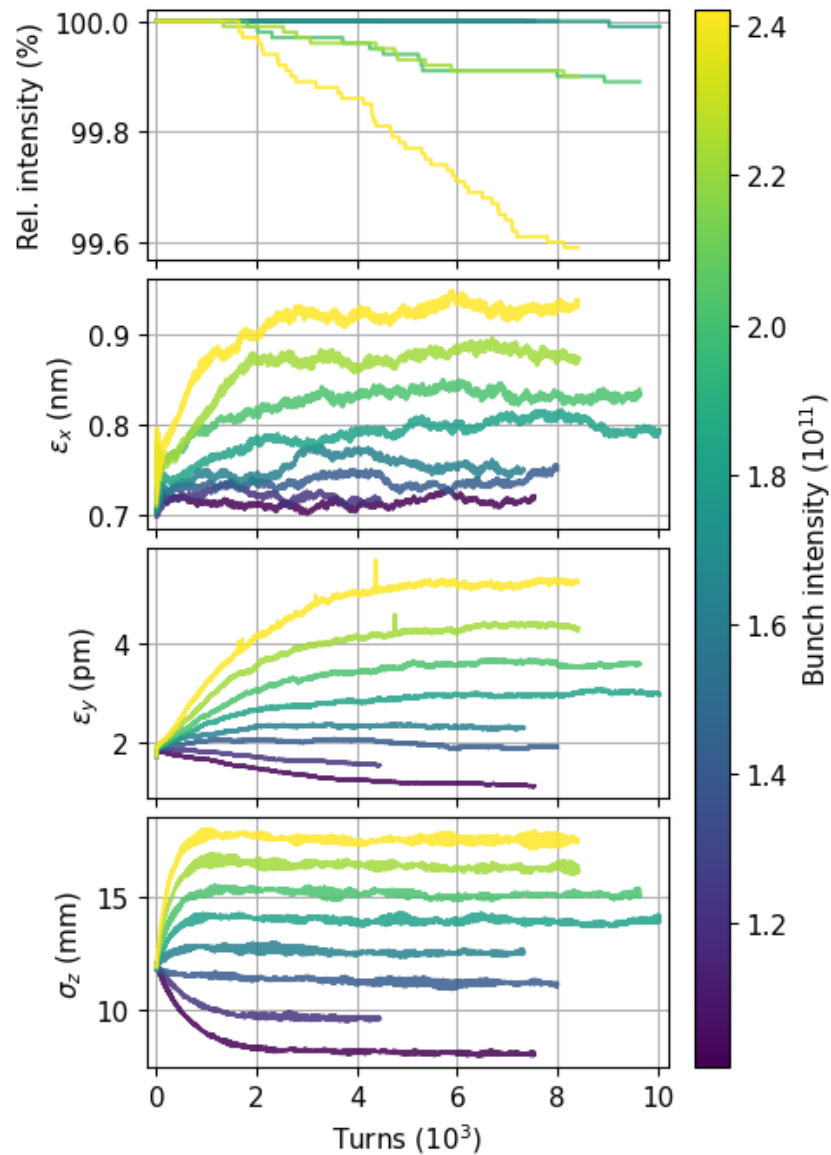


For a single seed with the tuning knobs + beam-beam, bunch intensity was scanned ($1 \times 10^{11} \rightarrow 2.4 \times 10^{11}$).

- At the nominal 2×10^{11} , $\epsilon_y \sim N^2$; below 1.4×10^{11} mostly linear

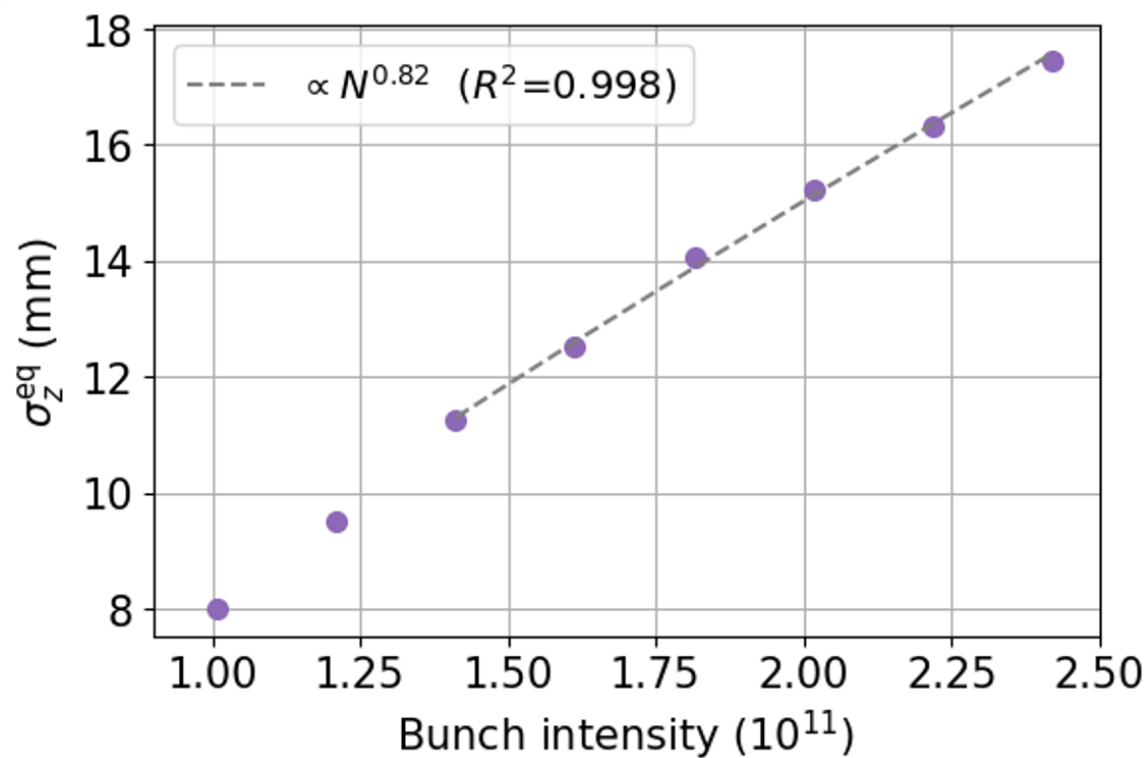


Probing the beam-beam limit

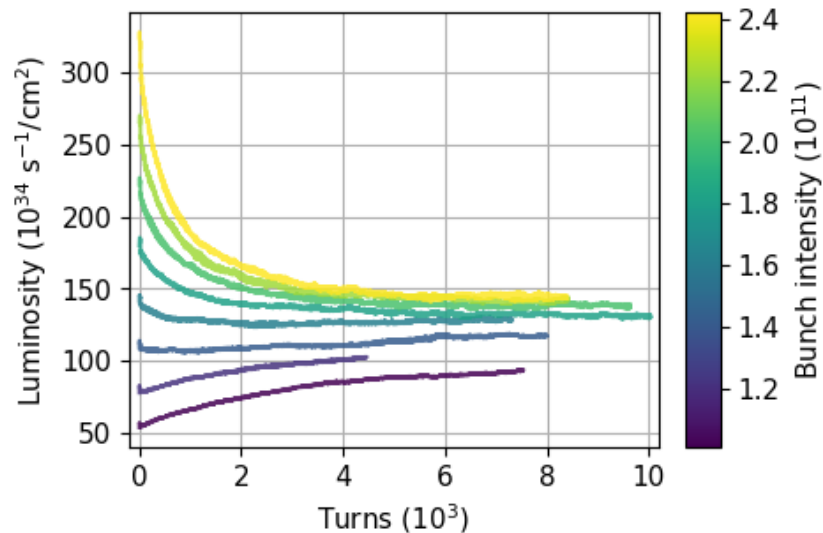


For a single seed with the tuning knobs + beam-beam, bunch intensity was scanned ($1 \times 10^{11} \rightarrow 2.4 \times 10^{11}$).

- At the nominal 2×10^{11} , $\epsilon_y \sim N^2$; below 1.4×10^{11} mostly linear
- Also, $\sigma_z \sim N$ (approximately?)

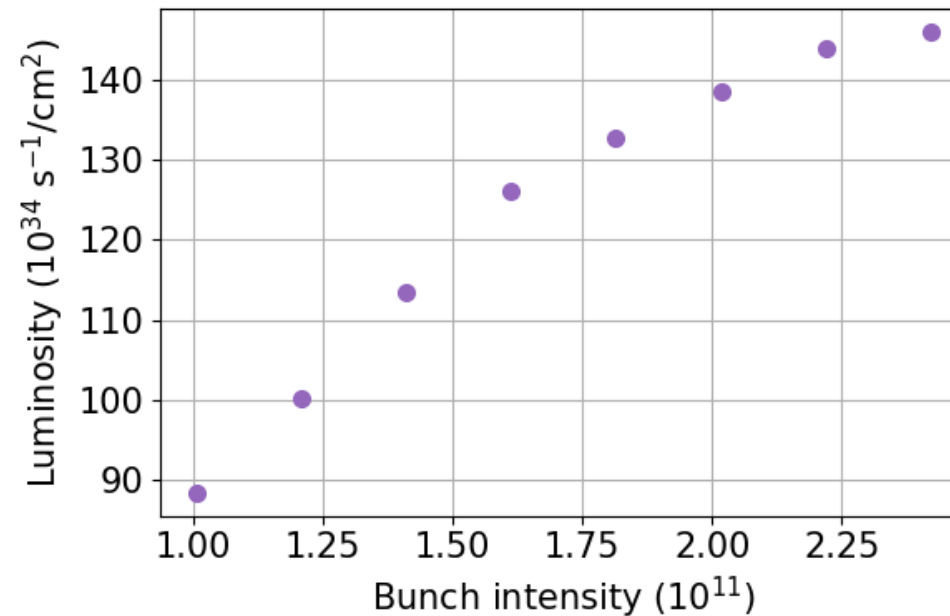


Probing the beam-beam limit



For a single seed with the tuning knobs + beam-beam, bunch intensity was scanned ($1 \times 10^{11} \rightarrow 2.4 \times 10^{11}$).

- At the nominal 2×10^{11} , $\epsilon_y \sim N^2$; below 1.4×10^{11} mostly linear
- Also, $\sigma_z \sim N$ (approximately?)
- Luminosity grows linearly and saturates (BS dominated?).



Luminosity margins should come from further optics improvements (even better control of coupling/dispersion, beam-beam rematching knob?).